

GEOPHYSICS

Ferroelastic hysteresis, shear modulus softening, and the tetragonal↔cubic transition in davemaoite

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Cubic CaSiO₃-perovskite (davemaoite), the third most abundant mineral in Earth's lower mantle, remains a key uncertainty in deep-Earth models. Using machine learning interatomic potentials and molecular dynamics simulations, we present a direct evidence of pronounced ferroelastic hysteresis near its tetragonal↔cubic transition. By isolating the intrinsic elastic response from anelastic relaxation and nonhydrostatic stress, we establish a robust upper-bound benchmark for sound velocities and elasticity models while revealing pronounced shear modulus softening across the transition. The confirmed ferroelasticity suggests substantial strain memory, seismic wave scattering, and strong attenuation if the transition is activated in regions of deeply subducted mid-ocean ridge basalt (MORB) accumulation. While pure CaPv may not undergo this transition at relevant conditions, it could occur in Ca(Si,Ti)O₃, contributing to the low shear velocity regions in the deep mantle and above the core-mantle boundary.

INTRODUCTION

The lower mantle, extending from the seismic discontinuity at ~660-km depth to the core-mantle boundary at ~2890 km, has been shown by seismic tomography models to contain large-scale velocity heterogeneities whose origins are still debated. Davemaoite, i.e., cubic CaSiO₃-perovskite (CaPv), comprises 7 to 10 vol % in pyrolite (1, 2) and up to 30 vol % in subducted mid-ocean ridge basalt (MORB) at depths below 560 km (3–5). As the third most abundant phase in the lower mantle, its presence significantly affects this region's properties (1) and those of subducted basaltic crust (3). Experimental (6) and theoretical (7–9) studies suggest that the solubility of CaPv in bridgmanite (MgSiO₃-perovskite), the most abundant mantle phase, increases with temperature and is completely dissolved at ~1800-km depth along the geotherm (10). Furthermore, CaPv is an important Ti-bearing phase in subducted slabs, forming a Ca(Si,Ti)O₃ solid solution (11, 12) with altered properties. It has been suggested that the key tetragonal↔cubic transition in Ti-bearing CaPv occurs at mid-mantle depths and may account for the seismic properties of large low shear velocity provinces (LLSVPs) (13). Despite the potential influence of the Ca(Si,Ti)O₃ and (Mg,Ca)SiO₃-perovskite solid solutions on deep mantle properties, the elastic behavior of pure CaPv under the mantle conditions remains highly relevant but debated and has yet to be fully addressed. Therefore, it is crucial to investigate its elastic properties first, particularly across the tetragonal↔cubic transition.

CaPv undergoes a tetragonal (*I4/mcm*) to cubic (*Pm3̄m*) phase transition with increasing temperature above ~12 GPa. Experimental (14, 15) and theoretical studies (16–18) of this transition have previously offered dissimilar phase boundaries. However, a recent large-scale computational study examining this phase boundary from

a thermodynamic viewpoint (19) confirms detailed measurements, pinning this transition at submantle-geotherm conditions. However, it remains unclear how this phase change may affect the elastic properties and acoustic velocities in CaPv at mantle conditions. The tetragonal↔cubic symmetry change in a perovskite system has long been predicted to exhibit ferroelastic behavior in the low-symmetry phase (20), characterized by non-Hookean deformation with a hysteresis loop, similar to other ferroic properties such as ferroelectricity and ferromagnetism. As CaPv belongs to the same perovskite family, one may anticipate that tetragonal CaPv also exhibits ferroelastic behavior. However, no complete ferroelastic hysteresis loop in tetragonal CaPv has been observed, unlike the one first identified in 1976 in the prototypical ferroelastic material Pb₃(PO₄)₂ (21). Furthermore, shear-wave velocity anomalies have been reported near ferroelastic transitions in several geophysically important materials, such as SiO₂ (22, 23). Because ferroelastic transitions vary in mechanism among systems, their associated elastic anomalies can differ in both form and the pressure-temperature range over which they occur. Unfortunately, the challenges posed by CaPv's spontaneous amorphization during decompression (24) have limited similar studies in this material. The struggle to capture such hysteresis loops and elastic anomalies in ultrasound measurements emphasizes the importance of accurate computational studies to understand the seismic properties of CaPv and similar ferroelastic systems. Here, we apply deep learning techniques to generate interatomic potentials that accurately model CaPv's compression curve and elastic properties at several temperatures. Exploring the hysteresis curve provides direct evidence of ferroelastic behavior in tetragonal CaPv. This method allows us to fully address the predicted yet previously unobserved shear modulus anomalies around the phase transition.

RESULTS

Ferroelastic behavior and hysteresis loop

We first assessed the accuracy of our trained deep learning potential (DP) model by comparing energies, forces, and virial terms using a validation dataset (fig. S1). The root mean square errors of the energy, force, and virial term are 3.6 meV/atom, 71 meV/Å, and 20 meV/atom, respectively. These uncertainties are comparable to

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those reported for other machine learning potentials (MLPs) under extreme conditions (19, 25–27). To further validate the model, we computed phonon dispersions using the phonon quasiparticle method (28, 29) to take the anharmonicity into consideration. The DP model accurately reproduces both harmonic and anharmonic phonon dispersions of CaPv (fig. S2), capturing the soft phonon modes at the R point in cubic CaPv at 0 K and their renormalization to positive frequencies at 3000 K due to the strong anharmonicity. The excellent agreement with such sensitive density functional theory (DFT) predictions highlights the DP model's reliability in describing lattice dynamics and its suitability for calculating elastic properties with comparable accuracy.

Molecular dynamics (MD) simulations using a deep learning potential trained with the strongly constrained and appropriately normed meta-generalized-gradient approximation functional (DFT-SCAN)

(30) can accurately reproduce the high-pressure elastic behavior of minerals, even across complex phase transitions (31). Comparison of our predicted 300 K DP-SCAN compression curve with experimental measurements (13, 32–36) is shown in Fig. 1A. We reproduce most of the measurements with unprecedented accuracy, with discrepancies smaller than 1%, except for data from Ricolleau *et al.* (37). This potential also reliably reproduces axial compressibility (Fig. 1B) and Si—O bond lengths (Fig. 1C). Notably, the Si—O bond length along the [001] direction remains slightly longer than along [100]. Compared to previous local density approximation (LDA) and generalized gradient approximation (GGA)-based predictions, SCAN reproduces these structural properties more accurately. Except for Ricolleau *et al.* (37) data, the compressive behavior of cubic CaPv measured by x-ray diffraction experiments at 2000 K was also well reproduced in our simulations up to 120 GPa (fig. S3). In addition,

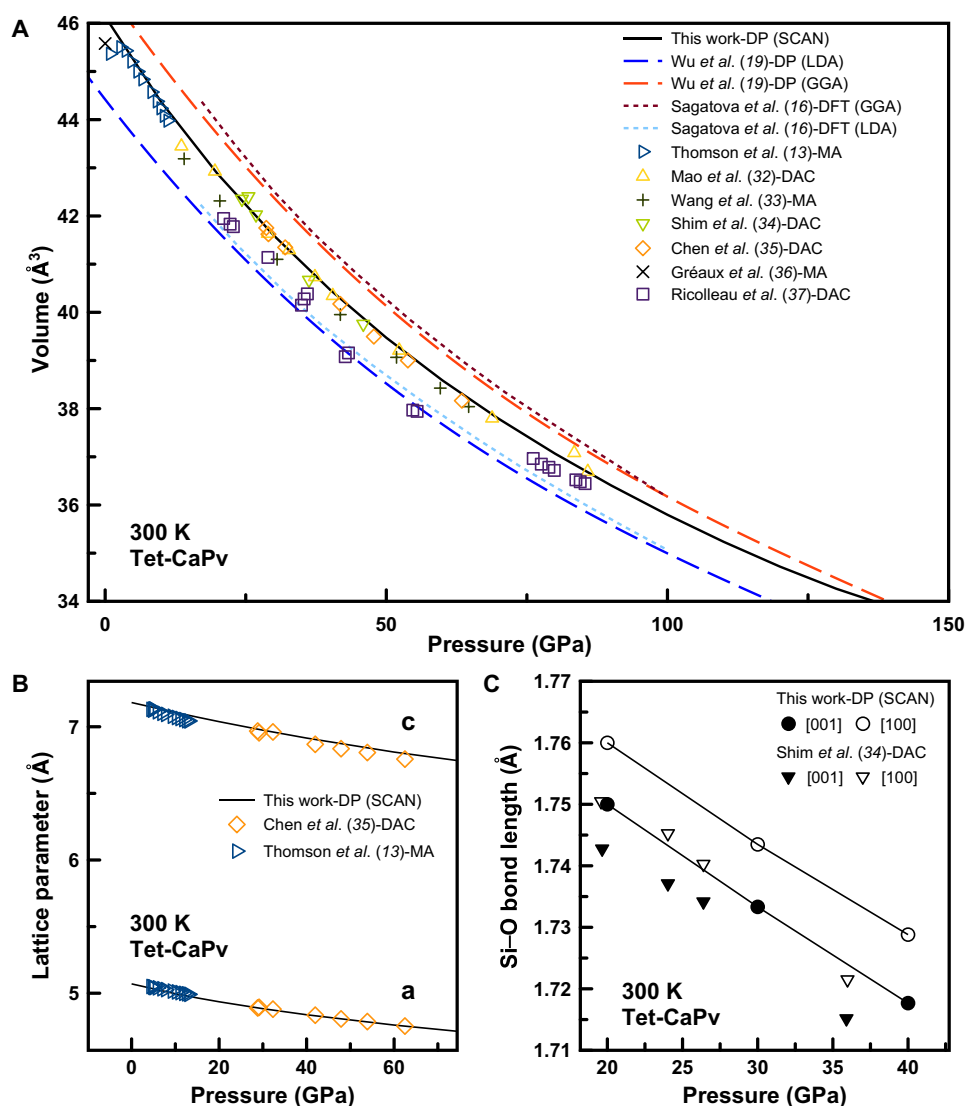


Fig. 1. Pressure dependence of the structural and compressional properties of tetragonal CaPv at 300 K. (A) Volume-pressure compression curves for tetragonal CaPv at 300 K. Solid curves are obtained using DP-SCAN, dashed lines are previous DFT results (16, 19), and experimental data are taken from (13, 32–37). (B) Lattice parameters of CaPv as a function of pressure at 300 K. (C) Si—O bond lengths in CaPv as a function of pressure at 300 K. Solid symbols denote Si—O bond along the c axis, while open symbols denote Si—O bond along a axes.

the thermal equation of state (EoS) parameters for cubic and tetragonal CaPv (table S1) and the isothermal EoS parameters for tetragonal CaPv (table S2) are consistent with both previous calculations and experimental measurements. Such agreement indicates that DP-SCAN describes the behavior of both tetragonal and cubic CaPv well at relevant conditions, validating the DP-SCAN method to explore the CaPv system throughout a broad range of conditions.

One of the key findings in this study is that tetragonal CaPv exhibits elastic hysteresis characteristic of ferroelastic systems under a stretch-compression cycle with uniaxial stress applied along the c axis (Fig. 2A). Such a feature was not seen in prior simulations or experiments whereas is consistent with the ferroelastic behavior observed in other perovskites (20). At 150 K and 12 GPa, compressive stress along the c axis causes the symmetry-breaking strain $e_t = [(c-a)/a]$ to approach 0, indicating a transition toward a

cubic-like structure. However, the cubic structure is unstable under these conditions, leading to a switch from a standing form (SF) to a lying-down form (LF) (Fig. 2A), corresponding to different orientations of the longer side of the tetragonal phase (text S2). Further analysis of the trajectories of oxygen ions reveals that SiO_6 octahedra rotate around the c axis in the SF state and around the $[110]$ direction in the LF state (Fig. 2, D and E). In the SF state, the tilting angle decreases with increasing compressive stress. In contrast, in the LF state, the tilting angle increases with compressive stress along the original c axis (Fig. 2F) since the compressive stress axis and the longer length axis are now perpendicular. The mechanical switching between these two states under external compressive stress results from a shear strain within the ab plane of the tetragonal cell, as illustrated in Fig. 2C. Microscopically, the system's stress versus strain relation still follows Hooke's law before and after the switching

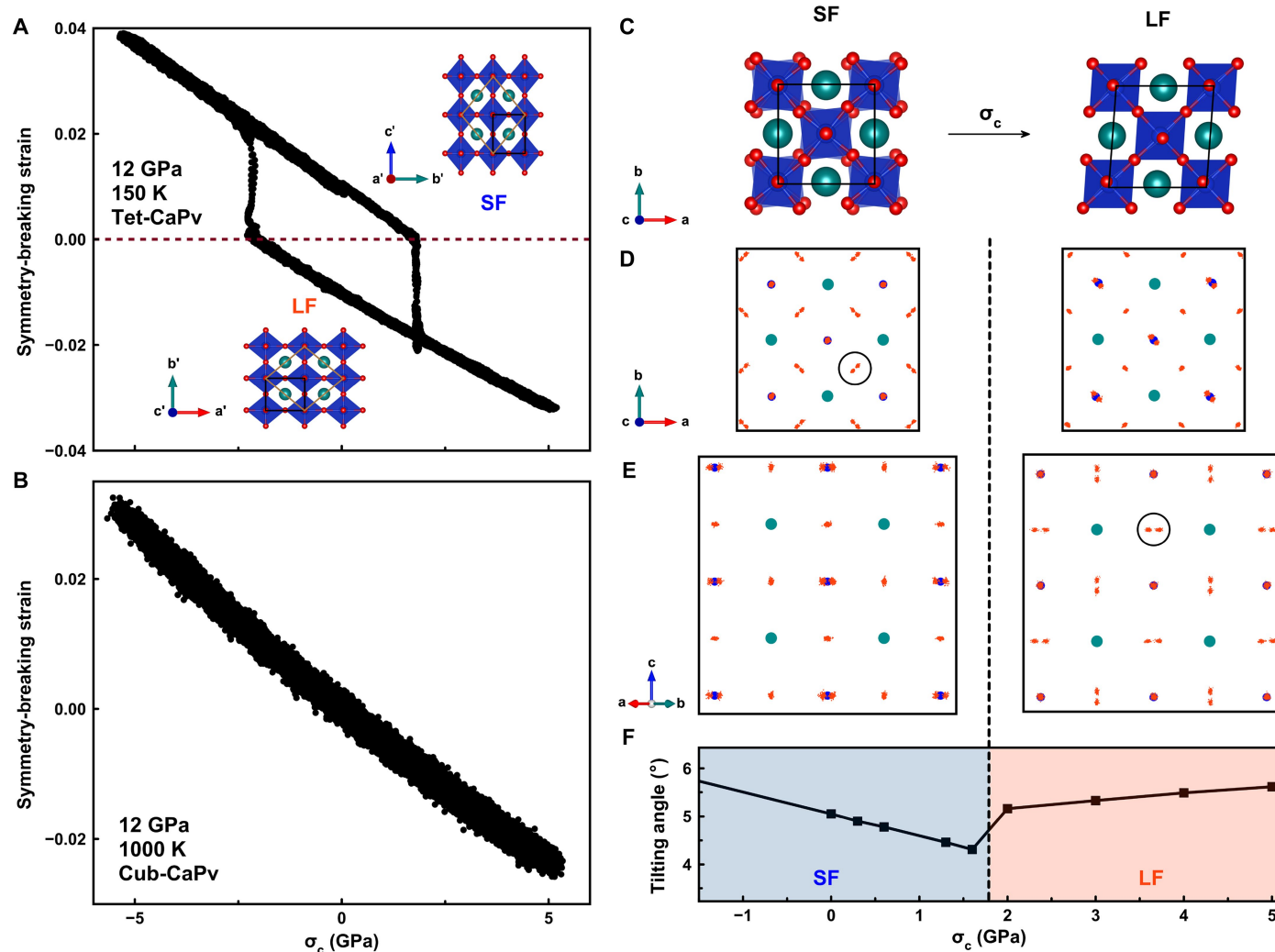


Fig. 2. Mechanical response and structural switching of CaPv under uniaxial stress. (A) Ferroelastic behavior with hysteresis loop in tetragonal CaPv at 12 GPa and 150 K. Black squares in the inset structures denote the corresponding cubic CaPv primitive cell ($1 \times 1 \times 1$, 5 atoms), while brown lozenge represents the tetragonal CaPv primitive cell ($\sqrt{2} \times \sqrt{2} \times 2$, 20 atoms). (a' , b' , c') represent the lattice vectors of the cubic cell, while (a , b , c) denote the lattice vectors of the tetragonal cell. (B) Paraelastic behavior in cubic CaPv at 12 GPa and 1000 K. Uniaxial stress was applied along c axis from -5 GPa (tensile) to 5 GPa (compression) gradually (text S2). (C) The crystal structure of tetragonal CaPv projected on the (001) plane of SF state and LF state. Blue, red, and dark cyan denote Si, O, and Ca ions. (D) The trajectories of O ions projected on the (001) plane. (E) The trajectories of O ions projected on the (110) plane. (F) SiO_6 octahedra tilting angles with respect to the longest axes in the SF and LF states.

(Fig. 2A). Also, the ferroelastic hysteresis loop's amplitude depends significantly on temperature. At 150 K and 12 GPa, the induced stress required to change the longer axis orientation is ~ 1.7 GPa. As the temperature approaches T_{tr} , the hysteresis loop shrinks and eventually disappears (fig. S5). This is the archetypical behavior of a ferroelastic material undergoing a transition from a low-symmetry ferroelastic phase to a high-symmetry paraelastic one (38), $I4/m\bar{c}m$ to $Pm\bar{3}m$ in this case. As shown in Fig. 2B, the ferroelastic hysteresis loop disappears entirely in cubic CaPv, the paraelastic counterpart. Ferroelastic $\leftrightarrow$ paraelastic transitions occur naturally in oxides and silicates within Earth's crust and mantle and have been reported to cause seismic velocity anomalies (39, 40). The effects of ferroelastic transitions on seismic parameters (22) are expected to differ significantly from those caused by other nonferroelastic structural transitions and temperature-compositional variations more commonly observed in the mantle (41). To better understand the nature of these transitions, we further investigate the complete set of elastic moduli across the transition and a broad range of pressures and temperatures around it.

Softening region near the phase transition

CaPv undergoes an improper ferroelastic $\leftrightarrow$ paraelastic transition at $T = T_{tr}$ similar to octahedral tilting perovskites systems such as SrTiO₃ (20). Individual components of the elastic tensors (Fig. 3) display strong anomalies near T_{tr} . The three sets of elastic coefficients for cubic CaPv, i.e., longitudinal ($c_{11} = c_{22} = c_{33}$), shear ($c_{44} = c_{55} = c_{66}$), and off-diagonal ($c_{12} = c_{13} = c_{23}$), gradually diverge at T_{tr} , indicating that the cubic lattice experiences increasingly anisotropic contraction and shear strains as temperature decreases. In tetragonal CaPv, the shear elastic coefficient c_{66} associated with shear strain ϵ_{xy} has a significantly larger value than the c_{44} or c_{55} coefficients. The smaller c_{44} and c_{55} values can be attributed to the rotation of SiO₆

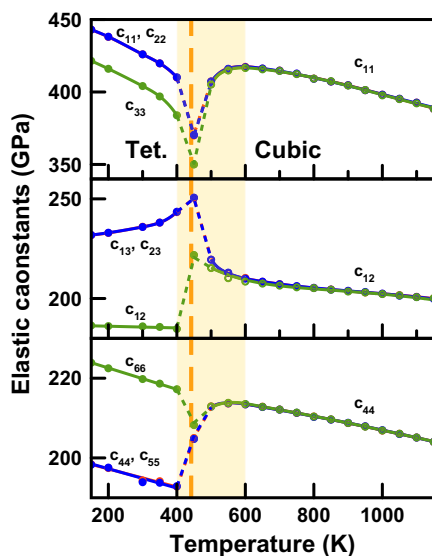


Fig. 3. Elastic constants versus temperature at 12 GPa obtained using the DP-SCAN potentials and methods listed under Materials and Methods. Orange dashed lines indicates the phase transition line measured by Thomson *et al.* (13). The dashed lines connecting the elastic coefficients of both phases to the minimum values are for guidance to the eye. The yellow-shaded area shows the “G-softening” region.

octahedra along the longer axis. This rotation would result from the lattice distortion transition where the cubic a axis of CaPv splits into the tetragonal a and c axes. The three principal longitudinal coefficients follow the trend $c_{11} > c_{33}$, which indicates anisotropic lattice distortions. The gradual symmetry reduction across the transition with a symmetry-breaking strain is usually accompanied by a marked anomaly of the shear modulus near T_{tr} .

To further evaluate CaPv's elastic properties across the ferroelastic $\leftrightarrow$ paraelastic transition, we analyzed the behavior of the adiabatic bulk modulus (B_s) and shear modulus (G) computed using Voigt-Reuss-Hill averages (Fig. 4, A and B). B_s aligns closely with previous measurements (13, 36) at 12 GPa and retains a nearly linear dependence in tetragonal and cubic CaPv away from the phase boundary. In addition, the modest reduction in B_s near T_{tr} at relatively low pressures is consistent with prior observations (fig. S6). However, our calculations reveal a strong, systematic dependence of this bulk modulus reduction (ΔB_s) when pressure increases beyond 60 GPa (fig. S7). Extreme compression of the crystal lattice inherently amplifies

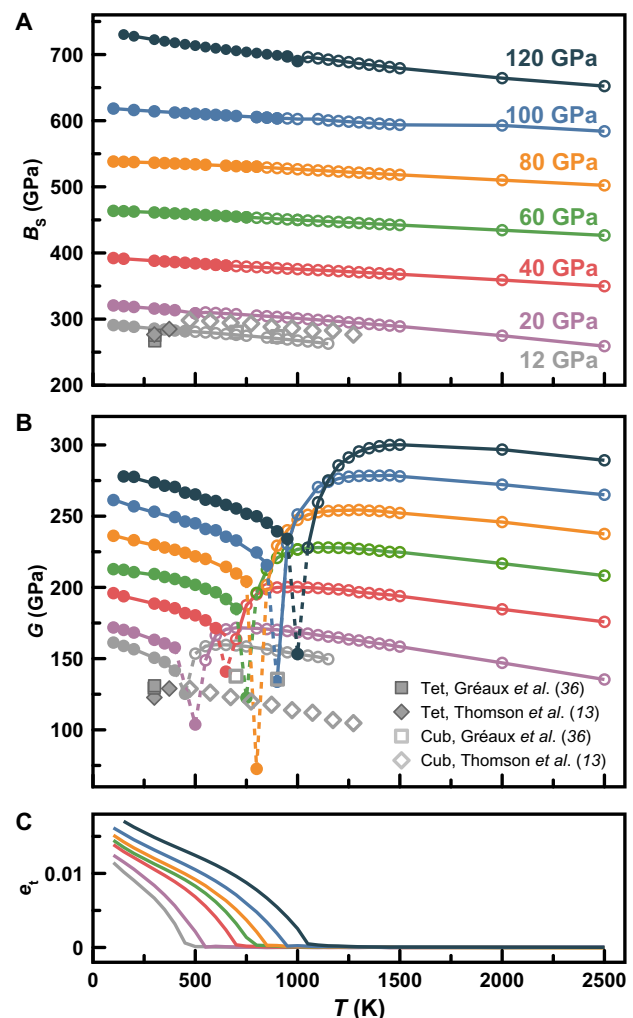


Fig. 4. Temperature and pressure evolution of the elastic moduli and structural strain in CaPv. (A) Bulk modulus- B_s , (B) shear modulus- G , and (C) symmetry-breaking strain of CaPv- ϵ_t as a function of temperature. Measured elastic moduli from previous studies are plotted as gray diamonds (13) (about 12 GPa) and gray squares (36) (12 ± 1 GPa).

the coupling between the volume strain and the octahedral tilting. Consequently, ΔB_s grows nonlinearly, reaching a much more pronounced reduction of ~ 2.1 GPa at 120 GPa. The discrepancies between our calculations and experimental measurements stem from different mechanisms depending on the structural phase. In the tetragonal phase, the less accentuated reduction of moduli observed in our calculations is primarily due to the absence of macroscopic domain structures and domain wall relaxations in our perfect single-crystal simulations. In the stable cubic phase, where ferroelastic domains are strictly absent, the measured and calculated elastic moduli are expected to converge. The remaining discrepancies in the cubic stability field likely arise from nonideal factors in the experimental samples, such as pure or defect-induced local strain fields that locally break cubic symmetry, unrelaxed deviatoric stresses inherent to high-pressure experimental assemblies or frequency-dependent anelastic relaxation observed in experiments, which are naturally excluded in our MD simulations. A recent similar simulation (42) does not show this small anomaly in B_s near T_{tr} , likely because it examined the elastic anomaly only in the high-pressure cubic phase. Notably, the transition does not cause significant changes in density (fig. S8) (43–45). In contrast to B_s , calculations reveal a significantly stronger reduction in G compared to measurements at 12 GPa. The characteristic dip in G occurs in the ferroelastic phase when the symmetry-breaking strain e_t approaches zero around the ferroelastic $\leftrightarrow$ paraelastic transition. An even stronger G softening occurs in the paraelastic phase near T_{tr} . Our predicted phase boundary occurs at a lower temperature compared to the latest computational estimate by Zhang *et al.* (42). We used different but equivalent approaches to outline the phase boundary—tracking elastic anomaly versus tracking lattice parameters in theirs [see Fig. 4 (B and C)]. Besides, our elastic properties agree well with theirs (fig. S9). Therefore, the difference in our phase boundaries should be primarily due to using different DFT functionals. Zhang *et al.* (42) used LDA with an approximate 4.3-GPa pressure shift, whereas we used SCAN, which produces the most accurate EoS to date (see figs. S1 and S2).

Determining the phase boundary of the tetragonal $\leftrightarrow$ cubic transition in ferroelastic CaPv has long been challenging due to the system's sensitivity to stress, pressure, and temperature, resulting in large uncertainties and discrepancies in reported boundaries (see Fig. 5A). Early computational studies in the mid-2000s used GGA-Perdew-Wang-1991 (GGA-PW91) (17, 46) and LDA (43) functionals, using either MD (17, 46) on relatively small supercells (~ 80 atoms) or free energy modeling based on Landau theory (43). These phase boundaries with large uncertainties suggested that the phase boundary could occur at subducting slabs or lower mantle conditions. Contemporaneous measurements performed on an externally laser-heated diamond anvil cell (LHDAC) (14, 15) offered much lower transition temperatures at mantle pressures. The tetragonal and cubic stability fields derived from these experiments were supported by DAC experiments in 2016 (47). In 2019, an elasticity-based determination of this boundary (13) indicated agreement with the previous measurements (14, 15) at ~ 12 GPa, but the extension/prediction of this phase boundary to mantle pressures based on ab initio MD on relatively small supercells (~ 80 atoms) using the GGA-Perdew–Burke–Ernzerhof (PBE) functional again suggested a phase boundary comparable to that determined using Landau theory (43). It was not until recently that MLPs trained on LDA and PBE/GGA using thermodynamic integration with MD (TIMD) simulations on large supercells 2004–2007 LHDAC measurements (14, 15) and constrained DFT

uncertainties within ~ 300 K throughout most of the mantle. Such a constrained boundary was also in good agreement with ab initio PBE calculations based on statistically converged quasi-harmonic approximation (QHA) calculations (16), a remarkable result given that CaPv is a highly anharmonic phase. A recent PBEsol calculation (48) using the stochastic self-consistent harmonic approximation yielded a phase boundary very similar to those determined using the LDA with TIMD (19) or PBE with QHA (16). These LDA/PBEsol/PBE-GGA statistically converged predictions, i.e., phonon-based with thorough Brillouin zone wave number sampling or MD-based large-scale (several thousand atoms) calculations, all point to lower temperature phase boundaries consistent with LHDAC experimental determinations (14, 15, 47). Alternative approaches, such as inferring the boundary from elastic softening (13, 42) or deviatoric stresses (48), have also been used. A recent MD-based LDA-trained MLP calculation of the elastic softening in cubic CaPv (42) suggested a phase boundary at intermediate temperatures, between the experimental ones (14, 15, 47), and the earlier ab initio MD (13) or Landau theory-based ones (43), which is still slightly high. Thus, their conclusion that the elastic softening in cubic CaPv induced by the transition could affect the seismic properties of subducting slabs, the D'' region, or LLSVPs, owing to the accumulation of CaPv-rich MORB in these regions, may not be entirely reliable for pure CaPv. However, the present large-scale SCAN-trained MLP MD simulations provide new constraints. First, it is statistically converged with respect to the number of atoms (10,240 atoms) and simulation run times (8 ns). Second, the SCAN density functional gives the most accurate thermal equations of state for tetragonal (300 K) and cubic CaPv (2000 K) phases, shown in Fig. 1 and fig. S3. Besides, the maximum elastic softening temperature (the phase boundary) at 12 GPa essentially coincides with that experimentally determined by Thomson *et al.* (13). The current phase boundary is also very close to those determined by statistically converged calculations using multiple DFT functionals and methods (16, 19, 48) and the experimental phase boundaries determined by LHDAC (14, 15, 47).

Notably, the shear modulus, G , exhibits a nearly linear temperature dependence well above and below T_{tr} but displays nonlinear behavior over a broad temperature range around T_{tr} (blue shaded area in Fig. 5B). The physical origin of this pronounced softening lies in intense local structural fluctuations near the phase transition point. As the temperature approaches T_{tr} , the lattice parameters a and c become nearly identical, and small thermal perturbations allow the system to dynamically fluctuate between the tetragonal and cubic structures. This dynamic structural instability strongly couples with macroscopic shear strain, leading to the pronounced dip in the shear modulus. Strictly speaking, the macroscopic ferroelastic behavior, characterized by structural hysteresis loops, is confined exclusively to the tetragonal phase and disappears entirely in the paraelastic cubic phase (Fig. 2B). However, the intrinsic structural effects of this ferroelastic transition persist well into the higher-temperature cubic stability field. Thus, we define the “softening region” (blue shaded area in Fig. 5B) as the temperature interval around the tetragonal-cubic transition where G deviates markedly from its nearly linear high- and low-temperature trends. This softening region spans 300 to 700 K, with the magnitude of softening strongly dependent on pressure. This broad softening interval contrasts with the narrow precursor softening (typically tens of kelvin) observed in analogous perovskite systems such as (Ca,Sr)TiO₃ at ambient pressure (49). Such extreme broadening is primarily driven by high pressure. At

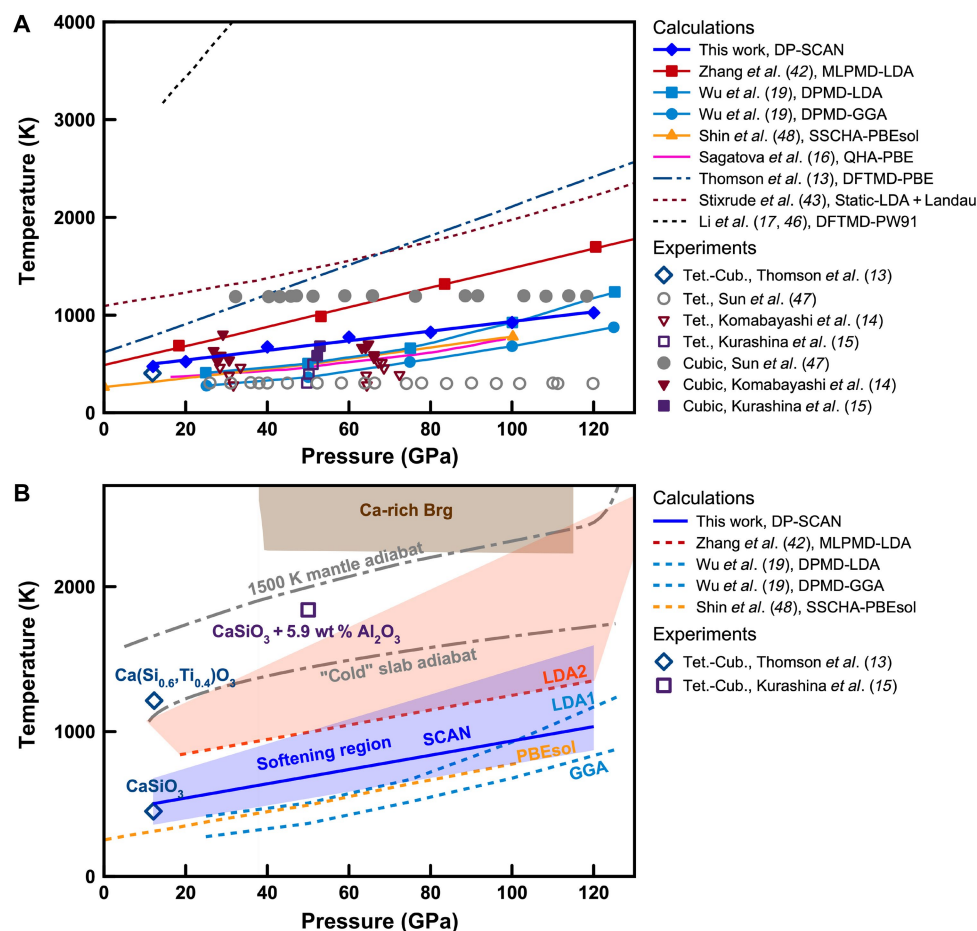


Fig. 5. Phase boundary and softening region of CaPv under lower-mantle conditions. (A) Tetragonal-cubic phase boundary of CaPv determined by experiments (13–15, 47) and theoretical calculations (16, 17, 19, 42, 43, 46, 48). (B) Softening region of CaPv. Normal mantle and cold slab geotherms (90) are plotted also. The blue line is the current DP-SCAN results corresponding to the mid-point between the minimum values of G in tetragonal and cubic CaPv in Fig. 4B. The blue-shaded area shows the G -softening region in this study, while the red-shaded area shows the results from Zhang *et al.* (42). Previous boundaries calculated using DP-GGA and DP-LDA (LDA1) (light blue dashed) (19), PBEsol (orange dashed) (48), and machine learning force fields LDA (LDA2) (red dashed) (42) are also shown. Dark blue diamonds represent measurements with different titanium concentrations (13). The brown-shaded region shows the observed stability field of Ca-rich bridgmanite (10).

12 GPa, the softening occurs over a narrower interval of ~ 300 K. With increasing pressure, the softening temperature range extensively broadens, reaching ~ 700 K at lower mantle conditions. Therefore, ferroelastic transitions can anomalously affect seismic parameters over a broad temperature range far beyond that of a sharp thermodynamic phase boundary. Furthermore, it is important to note that its exact width remains quantitatively sensitive to the chosen DFT functional and MLP architecture. For instance, a recent study (42) using a kernel-based Machine Learning Force Field (MLFF) trained on LDA reported a much wider precursor softening of ~ 400 K at 12 GPa. Although the G softening temperature range broadens with increasing pressure, it remains below the cold slab geotherm (Fig. 5B) where CaPv is expected to be abundant. This suggests that the significant softening associated with the tetragonal $\leftrightarrow$ cubic transition in pure CaPv alone is unlikely to cause seismic wave speed anomalies in the lower mantle. This result contrasts with previous elasticity results (42) that extend the softening region to the slab geotherm conditions (fig. S10). This discrepancy is due to their apparent overestimation of the tetragonal $\leftrightarrow$ cubic transition temperature (see Fig. 5B). Nonetheless, we do not rule out the possibility that certain young, fast, and

ultracold slabs, like Tonga (50), may meet the criteria for this transition, potentially giving rise to local seismic velocity anomalies.

Figure 6 shows cubic CaPv's elastic and seismic properties across various temperatures (T) and pressures (P). Figure 6B shows a typical behavior of B_S versus P and T , similar to bridgmanite's bulk modulus [e.g., Wan *et al.* (51)]. In contrast, G displays a strong non-linear P dependence at lower T s. For example, consistent with previous measurements (36), the G and V_S reduction at 1000 K are less marked versus P than versus T . At conditions where the elastic coefficients deviate significantly from nearly linear T dependence (Fig. 6A), G and wave velocities (Fig. 6, B and C) do as well. In summary, close to the ferroelastic $\leftrightarrow$ paraelastic transition (blue shaded area in Fig. 5B), the cross derivative, $\frac{\partial^2 D}{\partial T \partial P}$ (with D being an elastic coefficient, G , or wave velocity), displays clear T and P dependences.

DISCUSSION

Our work confronts a primary uncertainty in lower-mantle mineralogy: the elastic behavior of CaPv. Unlike MgSiO_3 , the most abundant

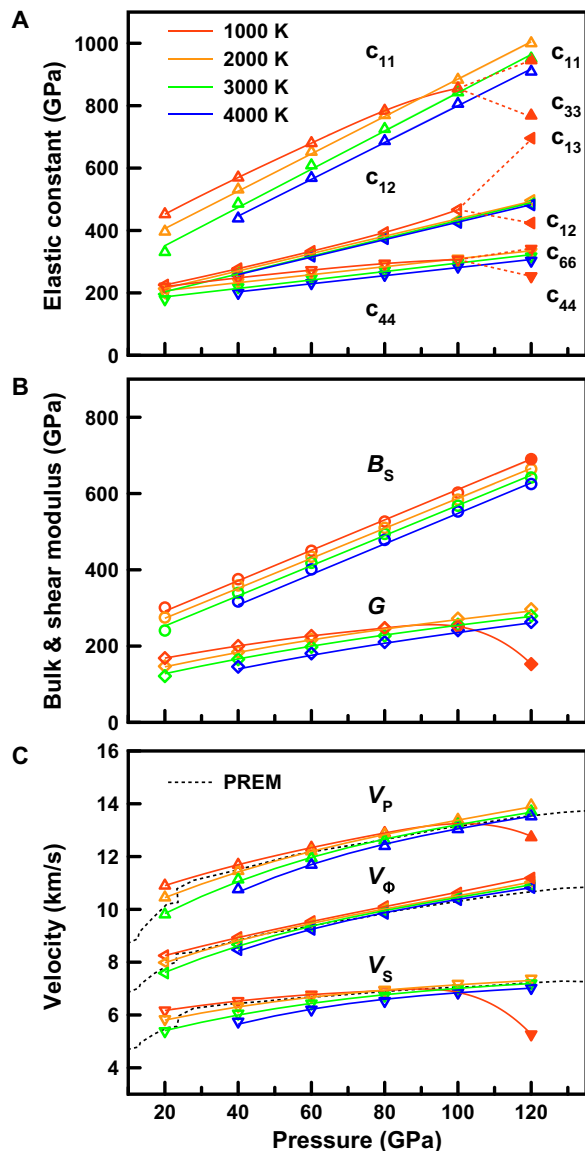


Fig. 6. Pressure dependence of adiabatic elastic constants, moduli, and seismic wave velocities of CaPv at high temperatures. (A) Adiabatic elastic constants of CaPv versus pressure at various temperatures calculated using stress-fluctuation formalism. Filled symbols denote the tetragonal CaPv, while open symbols denote the cubic CaPv. Fitting curves for each temperature are shown in lines. (B) Averaged bulk and shear moduli and (C) longitudinal (V_p), bulk sound (V_ϕ), and shear (V_s) wave velocities of CaPv at same temperatures. Black dashed lines indicate the PREM values (73).

mantle phase, which has been well characterized over the past decades (52–57), CaPv remains one of the largest unresolved uncertainties in lower-mantle mineral physics. Properly constraining this uncertainty requires recognition of the ferroelastic nature of CaPv's tetragonal $\leftrightarrow$ cubic transition and the hysteresis it introduces, which has been largely overlooked in previous studies (13, 19, 36, 58). By isolating the purely elastic response through MD simulations free from frequency-dependent relaxation and nonhydrostatic artifacts, we establish a conservative upper bound on seismic velocities in CaPv.

Ferroelastic systems are inherently complex and highly sensitive to external conditions, where even minor variations in stress, including chemical stress caused by alloying, can significantly alter the phase boundary, as well as the shape of elastic anomalies. Before this work, the ferroelastic behavior of CaPv had only been inferred indirectly, either by analogies with other ferroelastic minerals or through the observation of related elastic anomalies (20, 42). Such a ferroelastic system may appear cubic on average, but it can host coexisting domains of lower symmetry due to spontaneous symmetry-breaking strain fluctuations. Even a small stress can shift the system toward one orientation or another, giving rise to shear modulus softening. This sensitivity to stress and strain is illustrated by our calculation of the ferroelastic hysteresis loop, providing a predictive demonstration of strain memory associated with domain switching in a ferroelastic system.

Beyond the hysteresis itself, our simulations reveal a pronounced shear modulus anomaly that arises as an intrinsic precursor softening of CaPv, directly linked to the tetragonal $\leftrightarrow$ cubic transition. This dip exists even in the absence of domain wall relaxation or memory effects, making it a fundamental feature of the transition. This intrinsic softening corresponds to the same phenomenon highlighted on the cubic ($T > T_{tr}$) side in recent computational work (42), but here we capture the full transition and demonstrate its ferroelastic origin. Regarding inelasticity and seismic attenuation, our finite-size and finite-time MD calculations are designed to isolate the purely uniform elastic, unrelaxed high-frequency response, free from nonhydrostatic artifacts. Although slower macroscopic relaxation processes at seismic frequencies are not resolved directly, attenuation and velocity dispersion are strictly coupled, thereby reducing wave speeds at lower frequencies. Consequently, true seismic-frequency velocities can be significantly lower than the unrelaxed high-frequency values reported here, firmly establishing our purely elastic MD results as a conservative upper bound on seismic velocities. Within this intrinsic framework, the hysteresis loop reflects the system's strain memory, while the softening dip represents the fundamental elastic anomaly of the tetragonal $\leftrightarrow$ cubic transition. These two features together capture the complete elastic signature of ferroelasticity in CaPv, providing a baseline upon which further anelastic mechanisms, such as domain wall relaxation, act to drive the additional velocity reductions observed in experiments and in the real mantle.

Furthermore, nonhydrostatic stresses are expected to be present and significant in subducted MORB crust within descending plates or in D'' accumulations (59, 60). This issue is also highly relevant to experimental measurements at extreme pressures and temperatures, where the stress cannot be perfectly controlled. For instance, in experimental settings such as diamond anvil cell (DAC) measurements, differential stresses can be exceptionally large, reaching 5.3 to 7.2 GPa at bulk pressures around 145 GPa (61, 62). In the natural deep-Earth context, while macroscopic deviatoric stresses are generally expected to relax in hot mantle flow, the critical stress required to trigger the ferroelastic mechanical switching shrinks drastically as the temperature approaches the phase boundary (fig. S5). Consequently, even the remarkably small residual nonhydrostatic stresses sustained in deeply subducted slabs can profoundly influence ferroelastic domain populations and the apparent extent of elastic anomalies, provided that the slab geotherm intersects the phase boundary. Any additional stress-induced or anelastic effects associated with background deviatoric stress would further enhance seismic attenuation and reduce effective wave velocities relative to our intrinsic upper bound.

A further consequence of ferroelasticity is the strong frequency dependence (dispersion) of sound velocities near the ferroelastic $\leftrightarrow$ paraelastic transition (63, 64), which arises from strain-induced ferroelastic domain wall motion and the strain direction switching (49). At ultrasonic frequencies, twinned perovskites are considerably softer than predicted from calculations, as a portion of the applied stress can be accommodated by local spontaneous strain, as twin walls displace, rather than solely by a uniform and global crystal deformation (65). This relaxation mechanism reduces effective velocities and enhances attenuation of elastic waves. Domain wall motion amplifies elastic softening well beyond that produced by the phase transition itself (45). Because shear-wave velocities in CaPv are typically measured by ultrasound at 30 to 60 MHz, such domain wall dynamics provide a natural explanation for the consistent underestimation of the experimental shear modulus in the ferroelastic phase, and even in the paraelastic phase near T_{tr} , relative to the intrinsic upper-bound values reported here (13, 36). Therefore, as previously noted (36), CaPv near T_{tr} may exhibit even greater velocity reductions at seismic frequencies than those reported here. Similar behavior has been observed in LaCoO_3 , where twin wall motion results in pronounced attenuation—particularly at seismic-frequency and high-temperature conditions relevant to Earth's interior (66, 67).

However, domain wall attenuation alone cannot fully explain the sustained underestimation of shear wave speed and the 30% discrepancies between the measurements themselves away from the transition, up to $\sim 2 T_{tr}$. Additional contributors, such as deviatoric or unrelaxed stresses, local microstructural distortions, or other anelastic processes, are likely required to account for these discrepancies, specifically in the cubic phase where domain walls are absent. Furthermore, anelastic mechanisms beyond ferroelasticity—such as grain boundary sliding and creep—are known to affect seismic velocity but are exceedingly difficult to compute. Since dispersion in acoustic wave measurements always indicates anelasticity, targeted experimental investigations of anelastic mechanisms under these conditions are essential to clarify the origin of the differences between experimental and computational shear moduli, as well as discrepancies among measurements themselves. In addition to classical phonon softening, some perovskites exhibit central peak (relaxational) dynamics near the transition (e.g., LaAlO_3), which arise from disorder between local displacement states and can couple to acoustic modes, modifying the effective small- q elastic response (68). Our MD-based elasticity calculations and phonon-quasiparticle analysis capture anharmonic phonon renormalization and local fluctuations (fig. S2), but the finite supercell size and trajectory length limit the resolution at very small wave vectors and ultralow frequencies needed to quantify a central peak and the renormalized acoustic slope in the $q \rightarrow 0$ limit. Characterizing such contributions will require dedicated future calculations with enhanced small- q /low- ω resolution.

In this context, despite these inherent computational constraints on ultralow-frequency phenomena, the *ab initio* quality calculations presented here provide a necessary counterpoint: a robust and controlled means to isolate and quantify the elastic softening associated solely with ferroelasticity. Unlike experiments, they are free from frequency-dependent anelastic or nonhydrostatic artifacts and thus establish a conservative upper bound on seismic velocities. Combined with the dense sampling performed near the transition point, our results provide a critical benchmark for future investigations of ferroelastic behavior in CaPv. Together, our results indicate that the ferroelastic $\leftrightarrow$ paraelastic transition in CaPv, particularly

near T_{tr} , can act as a source of additional anelasticity. Such behavior would manifest as reduced velocities, enhanced attenuation, and seismic anomalies that might otherwise be interpreted as arising solely from temperature variations, compositional heterogeneities, or phase changes.

While pure CaPv does not introduce shear modulus anomalies under typical lower-mantle conditions via a pure elastic effect, the mineral likely incorporates substantial amounts of other elements. For example, since incompatible titanium (Ti) naturally enriches in MORB during partial melting of the upper mantle, CaPv acts as its primary host in the deep Earth (11). Previous studies indicate that CaPv in MORB can contain up to 28 wt % TiO_2 (3, 37, 69) or 4.8 wt % Al_2O_3 (3, 5, 70, 71) under lower-mantle conditions. This incorporation of other elements profoundly alters properties of CaPv through different mechanisms. First, as shown in Fig. 5B, the incorporation of Ti or Al stabilizes the low-symmetry phase and markedly shifts T_{tr} to higher temperatures than reported here, extending the influence of the tetragonal-cubic transition into the ambient mantle (12, 13, 15). Compositional complexity also broadens the softening interval. Analogous perovskite systems, such as $(\text{Ca,Sr})\text{TiO}_3$ solid solutions, exhibit massive acoustic attenuation and broadened shear softening over ~ 400 K, in contrast to the much narrow intervals of pure end-members (49). Such compositional broadening in Al_2O_3 -bearing CaPv within cold descending slabs may provide a plausible origin for certain seismic discontinuities observed in the lower mantle (72). In addition, the incorporation of Ti intrinsically reduces the acoustic wave velocities of CaPv significantly (69). While Wu's static calculations yield velocities that happen to align closely with the Preliminary Reference Earth Model (PREM; fig. S10) (73), these velocities will be substantially reduced under realistic high-temperature mantle conditions, causing the velocity profiles of Ti-bearing CaPv to drop significantly below PREM values. A recent study has also demonstrated that Fe,Al,Mg,Ti-bearing CaPv can exhibit a shear wave velocity 10 to 20% slower than pure CaPv (74). Consequently, the superposition of these effects firmly establishes that the moderate enrichment of MORB, characterized by high CaPv content and its capacity to host incompatible elements, provides a robust physical mechanism for the extreme low wave velocities characterizing LLSVPs (13, 74–76).

The influence of CaPv on the bulk elastic properties of the mantle must also be understood in the context of the CaSiO_3 - MgSiO_3 solid solution system. A recent prediction (9) and observation of the full (10) or partial (77) dissolution of CaPv in bridgmanite raise questions about how the seismic properties of the most abundant mantle phase, bridgmanite, might be affected by ferroelasticity in the CaPv endmember. Without a well-constrained understanding of the pure CaPv endmember, including its phase behavior and elastic response, any inference on the behavior of more complex solid solutions such as $\text{Ca}(\text{Si,Ti})\text{O}_3$ or $(\text{Ca,Mg})\text{SiO}_3$ risks being misleading or even fundamentally incorrect. While experimental measurements and computations have yet to advance to the point of accurately capturing or predicting fully dispersive acoustic wave speeds under geophysically relevant conditions, their interplay provides a deeper understanding of ferroelastic material properties. This interaction underscores the importance of integrating methodologies to address the nuanced complexities of ferroelasticity and its potential contributions to the overall anelastic behavior of minerals under the extreme conditions of the Earth's interior. In this regard, our results provide the essential benchmark and upper bound

against which all experimental and geophysical inferences of CaPv elasticity must be evaluated.

MATERIALS AND METHODS

Ab initio molecular dynamics (AIMD) simulations

To create benchmark datasets for CaPv, Born-Oppenheimer MD simulations were conducted using the Vienna Ab initio Simulation Package (78). The effectiveness of SCAN meta-GGA functional (30) was carefully investigated. The simulations used a $2 \times 2 \times 2$ supercell containing 40 atoms, alongside a $2 \times 2 \times 2$ k-point grid and a kinetic energy cutoff of 550 eV, which was shown to be sufficient to achieve convergence in calculating anharmonic phonon dispersions (79). One independent DP model, trained on SCAN datasets, was developed and is referred to as DP-SCAN. The model training used the smooth version descriptor `se_e2_a` proposed by Zhang *et al.* (80, 81) (text S1).

Deep potential MD simulations

We conducted deep potential MD simulations using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) code (82). A simulation was first performed in the Isothermal-isobaric ensemble (NPT) to equilibrate the cell shape under specific temperature and pressure conditions. This simulation involved 10,240 atoms, ran for 0.1 ns, and used a time step of 0.5 fs. The Nosé-Hoover thermo-barostat that incorporates modular invariance (83, 84) was applied for temperature and pressure control. After the system reached equilibrium, subsequent simulations started from the equilibrated unit cell shape. To determine the elastic tensor through the stress-fluctuation approach, simulations were first conducted in the Canonical ensemble (NVT) to gather atomic coordinates and velocities at target temperatures. These were followed by the 8-ns-long simulations in the Microcanonical ensemble (NVE) with a time step of 2 fs to compute the adiabatic $\mathbf{c}_{ij}^s$ tensor components under defined conditions.

The stress-fluctuation formalism for thermoelasticity

The thermodynamic elastic tensor $A_{ijkl}(T, V)$ (where $i, j, k, l = 1, 2, 3$) represents the second derivative of the free energy F density with respect to strain under equilibrium conditions, while the volume V is fixed in MD simulations. It is expressed as (85–87)

$$A_{ijkl}(T, V) = \frac{1}{V} \frac{\partial^2 F}{\partial \epsilon_{ij} \partial \epsilon_{kl}} = \langle A_{ijkl}^B \rangle - \frac{k_B T}{V} [\langle \sigma_{ij} \sigma_{kl} \rangle - \langle \sigma_{ij} \rangle \langle \sigma_{kl} \rangle]$$

where $\langle \dots \rangle$ denotes the ensemble average, ϵ_{ij} is the strain tensor, N is the number of atoms, T is the temperature, and δ_{ij} is the Kronecker delta function. The stress tensor σ_{ij} and Born matrix tensor A_{ijkl}^B are derived from the first and second derivatives of strain energy density U/V with respect to ϵ_{ij} , respectively

$$\sigma_{ij} = \frac{1}{V} \frac{\partial U}{\partial \epsilon_{ij}}, \quad A_{ijkl}^B = \frac{1}{V} \frac{\partial^2 U}{\partial \epsilon_{ij} \partial \epsilon_{kl}}$$

During the equilibrated MD runs, both A_{ijkl}^B and σ_{ij} are periodically calculated. Since the convergence of A_{ijkl}^B is faster than that of σ_{ij} , different sampling frequencies are used to optimize storage while maintaining computational accuracy (calculating A_{ijkl}^B every 200 MD steps and σ_{ij} every 20 MD steps). The Born matrix A_{ijkl}^B is computed as follows (87)

$$A_{ijkl}^B = \frac{\partial \sigma_{ij}}{\partial \epsilon_{kl}} = \sigma_{ij} \delta_{kl} + \sigma_{ik} \delta_{jl}$$

For an average stress σ_{ij} calculated during the simulation, the effective elastic tensor C_{ijkl} is given by the following expression (88, 89)

$$C_{ijkl} = \frac{1}{V} \frac{\partial^2 F}{\partial \epsilon_{ij} \partial \epsilon_{kl}} - \sigma_{ij} \delta_{kl} + \frac{1}{2} (\sigma_{ik} \delta_{jl} + \sigma_{kj} \delta_{il} + \sigma_{il} \delta_{jk} + \sigma_{lj} \delta_{ik})$$

Under hydrostatic stress condition [i.e., $\sigma_{ij} = -P\delta_{ij}$, where $P = -\frac{1}{3} \text{Tr}(\sigma_{ij})$], this equation can be further simplified to (88)

$$C_{ijkl} = \frac{1}{V} \frac{\partial^2 F}{\partial \epsilon_{ij} \partial \epsilon_{kl}} + P(\delta_{ij} \delta_{kl} - \delta_{il} \delta_{kj} - \delta_{ik} \delta_{jl})$$

Here, A_{ijkl} and C_{ijkl} are equal only when the system is free from external pressure or stress. The adiabatic elastic tensors A_{ijkl}^S and C_{ijkl}^S can be obtained from NVE simulations.

Supplementary Materials

This PDF file includes:

Supplementary Text S1 and S2

Figs. S1 to S11

Tables S1 and S2

References

REFERENCES

1. T. Irifune, Absence of an aluminous phase in the upper part of the Earth's lower mantle. *Nature* **370**, 131–133 (1994).
2. A. Saikia, D. J. Frost, D. C. Rubie, Splitting of the 520-kilometer seismic discontinuity and chemical heterogeneity in the mantle. *Science* **319**, 1515–1518 (2008).
3. K. Hirose, Y. Fei, Subsolidus and melting phase relations of basaltic composition in the uppermost lower mantle. *Geochim. Cosmochim. Acta* **66**, 2099–2108 (2002).
4. T. Irifune, A. E. Ringwood, Phase transformations in subducted oceanic crust and buoyancy relationships at depths of 600–800 km in the mantle. *Earth Planet. Sci. Lett.* **117**, 101–110 (1993).
5. S. E. Kesson, J. D. Fitz Gerald, J. M. G. Shelley, Mineral chemistry and density of subducted basaltic crust at lower-mantle pressures. *Nature* **372**, 767–769 (1994).
6. T. Irifune, M. Miyashita, T. Inoue, J. Ando, K. Funakoshi, W. Utsumi, High-pressure phase transformation in CaMgSi₂O₆ and implications for origin of ultra-deep diamond inclusions. *Geophys. Res. Lett.* **27**, 3541–3544 (2000).
7. L. Vitos, B. Magyari-Köpe, R. Ahuja, J. Kollár, G. Grimvall, B. Johansson, Phase transformations between garnet and perovskite phases in the Earth's mantle: A theoretical study. *Phys. Earth Planet. In.* **156**, 108–116 (2006).
8. D. Y. Jung, M. W. Schmidt, Solid solution behaviour of CaSiO₃ and MgSiO₃ perovskites. *Phys. Chem. Miner.* **38**, 311–319 (2011).
9. J. M. R. Muir, A. R. Thomson, F. Zhang, The miscibility of calcium silicate perovskite and bridgmanite: A single perovskite solid solution in hot, iron-rich regions. *Earth Planet. Sci. Lett.* **566**, 116973–116973 (2021).
10. B. Ko, E. Greenberg, V. Prakapenka, E. E. Alp, W. Bi, Y. Meng, D. Zhang, S.-H. Shim, Calcium dissolution in bridgmanite in the Earth's deep mantle. *Nature* **611**, 88–92 (2022).
11. K.-H. Chao, M. Berrada, S. Wang, J. Peckenpaugh, D. Zhang, S. Chariton, V. Prakapenka, B. Chen, Structure and equation of state of Ti-bearing davemaoite: New insights into the chemical heterogeneity in the lower mantle. *Am. Mineral.* **109**, 1861–1870 (2024).
12. A. R. Thomson, M. J. Walter, S. C. Kohn, R. A. Brooker, Slab melting as a barrier to deep carbon subduction. *Nature* **529**, 76–79 (2016).
13. A. R. Thomson, W. A. Crichton, J. P. Brodholt, I. G. Wood, N. C. Siersch, J. M. R. Muir, D. P. Dobson, S. A. Hunt, Seismic velocities of CaSiO₃ perovskite can explain LLSVPs in Earth's lower mantle. *Nature* **572**, 643–647 (2019).
14. T. Komabayashi, K. Hirose, N. Sata, Y. Ohishi, L. S. Dubrovinsky, Phase transition in CaSiO₃ perovskite. *Earth Planet. Sci. Lett.* **260**, 564–569 (2007).
15. T. Kurashina, K. Hirose, S. Ono, N. Sata, Y. Ohishi, Phase transition in Al-bearing CaSiO₃ perovskite: Implications for seismic discontinuities in the lower mantle. *Phys. Earth Planet. In.* **145**, 67–74 (2004).

16. D. N. Sagatova, A. F. Shatskiy, N. E. Sagatov, K. D. Litasov, Phase Relations in CaSiO_3 System up to 100 GPa and 2500 K. *Geochim. Int.* **59**, 791–800 (2021).
17. L. Li, D. J. Weidner, J. Brodholt, D. Alfé, G. David Price, R. Caracas, R. Wentzcovitch, Elasticity of CaSiO_3 perovskite at high pressure and high temperature. *Phys. Earth Planet. In.* **155**, 249–259 (2006).
18. R. Caracas, R. Wentzcovitch, G. D. Price, J. Brodholt, CaSiO_3 perovskite at lower mantle pressures. *Geophys. Res. Lett.* **32**, 1–5 (2005).
19. F. Wu, Y. Sun, T. Wan, S. Wu, R. M. Wentzcovitch, Deep-learning-based prediction of the tetragonal $\rightarrow$ cubic transition in davemaoite. *Geophys. Res. Lett.* **51**, e2023GL108012 (2024).
20. M. A. Carpenter, E. K. H. Salje, Elastic anomalies in minerals due to structural phase transitions. *Eur. J. Mineral.* **10**, 693–812 (1998).
21. E. Salje, G. Hoppmann, Direct observation of ferroelasticity in $\text{Pb}_3(\text{PO}_4)_2 \cdot \text{Pb}_3(\text{VO}_4)_2$. *Mater. Res. Bull.* **11**, 1545–1549 (1976).
22. Y. Zhang, S. Fu, B. Wang, J. F. Lin, Elasticity of a pseudoproper ferroelastic transition from stishovite to post-stishovite at high pressure. *Phys. Rev. Lett.* **126**, 025701–025701 (2021).
23. M. A. Carpenter, R. J. Hemley, H. K. Mao, High-pressure elasticity of stishovite and the $\text{P42/mnm} \rightleftharpoons \text{Pnnm}$ phase transition. *J. Geophys. Res. Solid Earth* **105**, 10807–10816 (2000).
24. L. G. Liu, A. E. Ringwood, Synthesis of a perovskite-type polymorph of CaSiO_3 . *Earth Planet. Sci. Lett.* **28**, 209–211 (1975).
25. J. Deng, L. Stixrude, Thermal conductivity of silicate liquid determined by machine learning potentials. *Geophys. Res. Lett.* **48**, e2021GL093806 (2021).
26. Z. Zhang, G. Csányi, D. Alfé, Partitioning of sulfur between solid and liquid iron under Earth's core conditions: Constraints from atomistic simulations with machine learning potentials. *Geochim. Cosmochim. Acta* **291**, 5–18 (2020).
27. F. Xu, D. Wang, Z. Li, H. Song, L. Liu, H. Geng, J. Hu, X. Chen, Large-scale simulation of thermal conductivity in CaSiO_3 perovskite with neuroevolution potential. *Appl. Phys. Lett.* **125**, 5–5 (2024).
28. Z. Zhang, D. B. Zhang, T. Sun, R. M. Wentzcovitch, phq: A Fortran code to compute phonon quasiparticle properties and dispersions. *Comput. Phys. Commun.* **243**, 110–120 (2019).
29. D. B. Zhang, T. Sun, R. M. Wentzcovitch, Phonon quasiparticles and anharmonic free energy in complex systems. *Phys. Rev. Lett.* **112**, 058501 (2014).
30. J. Sun, A. Ruzsinszky, J. P. Perdew, Strongly constrained and appropriately normed semilocal density functional. *Phys. Rev. Lett.* **115**, 036402 (2015).
31. C. Luo, Y. Sun, R. Wentzcovitch, Elasticity and acoustic velocities of δ - AlOOH at extreme conditions: A methodology assessment. *Phys. Rev. Mater.* **8**, 103601 (2024).
32. H. K. Mao, L. C. Chen, R. J. Hemley, A. P. Jephcoat, Y. Wu, W. A. Bassett, Stability and equation of state of CaSiO_3 -perovskite to 134 GPa. *J. Geophys. Res.* **94**, 17889–17894 (1989).
33. Y. Wang, D. J. Weidner, F. Guyot, Thermal equation of state of CaSiO_3 perovskite. *J. Geophys. Res. Solid Earth* **101**, 661–672 (1996).
34. S. H. Shim, R. Jeanloz, T. S. Duffy, Tetragonal structure of CaSiO_3 perovskite above 20 GPa. *Geophys. Res. Lett.* **29**, 2166–2166 (2002).
35. H. Chen, S. H. Shim, K. Leinenweber, V. Prakapenka, Y. Meng, C. Prescher, Crystal structure of CaSiO_3 perovskite at 28–62 GPa and 300 K under quasi-hydrostatic stress conditions. *Am. Mineral.* **103**, 462–468 (2018).
36. S. Gréaux, T. Irifune, Y. Higo, Y. Tange, T. Arimoto, Z. Liu, A. Yamada, Sound velocity of CaSiO_3 perovskite suggests the presence of basaltic crust in the Earth's lower mantle. *Nature* **565**, 218–221 (2019).
37. A. Ricolleau, J. P. Perrillat, G. Fiquet, I. Daniel, J. Matas, A. Addad, N. Menguy, H. Cardon, M. Mezouar, N. Guignot, Phase relations and equation of state of a natural MORB: Implications for the density profile of subducted oceanic crust in the Earth's lower mantle. *J. Geophys. Res. Solid Earth* **115**, B08202 (2010).
38. E. K. H. Salje, Ferroelastic materials. *Annu. Rev. Mat. Res.* **42**, 265–283 (2012).
39. S. Kaneshima, Seismic scatterers in the mid-lower mantle. *Phys. Earth Planet. In.* **257**, 105–114 (2016).
40. E. K. H. Salje, Application of Landau theory for the analysis of phase transitions in minerals. *Phys. Rep.* **215**, 49–99 (1992).
41. J. Zhuang, R. Wentzcovitch, Fe^{2+} partitioning in al-free pyrolite: Consequences for seismic velocities and heterogeneities. *Geophys. Res. Lett.* **51**, e2024GL108967 (2024).
42. C. Zhang, J.-Y. Yang, T. Sun, H. Zhang, J. P. Brodholt, Strong precursor softening in cubic CaSiO_3 perovskite. *Proc. Natl. Acad. Sci. U.S.A.* **122**, e2410910122 (2025).
43. L. Stixrude, C. Lithgow-Bertelloni, B. Kiefer, P. Fumagalli, Phase stability and shear softening in CaSiO_3 perovskite at high pressure. *Phys. Rev. B* **75**, 024108 (2007).
44. T. Tsuchiya, Elasticity of subducted basaltic crust at the lower mantle pressures: Insights on the nature of deep mantle heterogeneity. *Phys. Earth Planet. In.* **188**, 142–149 (2011).
45. M. A. Carpenter, Elastic anomalies accompanying phase transitions in $(\text{Ca},\text{Sr})\text{TiO}_3$ perovskites: Part I. Landau theory and a calibration for SrTiO_3 . *Am. Mineral.* **92**, 309–327 (2007).
46. L. Li, D. J. Weidner, J. Brodholt, D. Alfé, G. D. Price, R. Caracas, R. Wentzcovitch, Phase stability of CaSiO_3 perovskite at high pressure and temperature: Insights from ab initio molecular dynamics. *Phys. Earth Planet. In.* **155**, 260–268 (2006).
47. N. Sun, Z. Mao, S. Yan, X. Wu, V. B. Prakapenka, J. F. Lin, Confirming a pyrolytic lower mantle using self-consistent pressure scales and new constraints on CaSiO_3 perovskite. *J. Geophys. Res. Solid Earth* **121**, 4876–4894 (2016).
48. Y. Shin, E. Di Lucente, N. Marzari, L. Monacelli, The thermodynamics of CaSiO_3 perovskite in Earth's lower mantle. *Phys. Rev. B* **112**, 174113 (2025).
49. N. J. Perks, Z. Zhang, R. J. Harrison, M. A. Carpenter, Strain relaxation mechanisms of elastic softening and twin wall freezing associated with structural phase transitions in $(\text{Ca},\text{Sr})\text{TiO}_3$ perovskites. *J. Phys. Condens. Matter* **26**, 505402 (2014).
50. E. M. Syracuse, P. E. van Keken, G. A. Abers, D. Suetsugu, C. Bina, T. Inoue, D. Wiens, M. Jellinek, The global range of subduction zone thermal models. *Phys. Earth Planet. In.* **183**, 73–90 (2010).
51. T. Wan, C. Luo, Y. Sun, R. M. Wentzcovitch, Thermoelastic properties of bridgmanite using deep-potential molecular dynamics. *Phys. Rev. B* **109**, 094101 (2024).
52. A. Yeganeh-Haeri, Synthesis and re-investigation of the elastic properties of single-crystal magnesium silicate perovskite. *Phys. Earth Planet. In.* **87**, 111–121 (1994).
53. S. V. Sinogeikin, J. Zhang, J. D. Bass, Elasticity of single crystal and polycrystalline MgSiO_3 perovskite by Brillouin spectroscopy. *Geophys. Res. Lett.* **31**, e2004GL019559 (2004).
54. G. Criniti, A. Kurnosov, T. Boffa Ballaran, D. J. Frost, Single-crystal elasticity of MgSiO_3 bridgmanite to mid-lower mantle pressure. *J. Geophys. Res. Solid Earth* **126**, e022020JB020967 (2021).
55. R. M. Wentzcovitch, B. B. Karki, M. Cococcioni, S. de Gironcoli, Thermoelastic properties of MgSiO_3 -perovskite: Insights on the nature of the Earth's lower mantle. *Phys. Rev. Lett.* **92**, 018501 (2004).
56. A. R. Oganov, J. P. Brodholt, G. D. Price, Ab initio elasticity and thermal equation of state of MgSiO_3 perovskite. *Earth Planet. Sci. Lett.* **184**, 555–560 (2001).
57. B. Wehinger, A. Bosak, S. Nazzareni, D. Antonangeli, A. Mirone, S. L. Chaplot, R. Mittal, E. Ohtani, A. Shatskiy, S. Saxena, S. Ghose, M. Krisch, Dynamical and elastic properties of MgSiO_3 perovskite (bridgmanite). *Geophys. Res. Lett.* **43**, 2568–2575 (2016).
58. K. Kawai, T. Tsuchiya, Small shear modulus of cubic CaSiO_3 perovskite. *Geophys. Res. Lett.* **42**, 2718–2726 (2015).
59. A. Nowacki, J. Wookey, J. M. Kendall, Deformation of the lowermost mantle from seismic anisotropy. *Nature* **467**, 1091–1094 (2010).
60. M. D. Long, Complex anisotropy in D'' beneath the eastern Pacific from SKS-SKKS splitting discrepancies. *Earth Planet. Sci. Lett.* **283**, 181–189 (2009).
61. L. Miyagi, W. Kaniyacharoen, P. Kaercher, K. K. M. Lee, H.-R. Wenk, Slip systems in MgSiO_3 post-perovskite: Implications for D'' anisotropy. *Science* **329**, 1639–1641 (2010).
62. S. B. Merkel, A. K. McNamara, A. Kubo, S. Speziale, L. Miyagi, Y. Meng, T. S. Duffy, H.-R. Wenk, Deformation of $(\text{Mg},\text{Fe})\text{SiO}_3$ post-perovskite and D'' anisotropy. *Science* **316**, 1729–1732 (2007).
63. A. Kohutych, R. Yevych, S. Perechinskii, V. Samulionis, J. Banys, Y. Vysochanskii, Sound behavior near the Lifshitz point in proper ferroelectrics. *Phys. Rev. B* **82**, 054101 (2010).
64. R. J. Harrison, S. A. T. Redfern, J. Street, The effect of transformation twins on the seismic-frequency mechanical properties of polycrystalline $\text{Ca}_{1-x}\text{Sr}_x\text{TiO}_3$ perovskite. *Am. Mineral.* **88**, 574–582 (2003).
65. M. A. Carpenter, Elastic properties of minerals and the influence of phase transitions. *Am. Mineral.* **91**, 229–246 (2006).
66. M. A. Carpenter, Z. Zhang, Anelasticity maps for acoustic dissipation associated with phase transitions in minerals. *Geophys. J. Int.* **186**, 279–295 (2011).
67. Z. Zhang, J. Koppensteiner, W. Schranz, D. Prabhakaran, M. A. Carpenter, Strain coupling mechanisms and elastic relaxation associated with spin state transitions in LaCoO_3 . *J. Phys. Condens. Matter* **23**, 145401 (2011).
68. M. A. Carpenter, S. V. Sinogeikin, J. D. Bass, Elastic relaxations associated with the $\text{Pm}\bar{3}\text{m} \rightarrow \text{R}\bar{3}\text{m}$ transition in LaAlO_3 : II. Mechanisms of static and dynamical softening. *J. Phys. Condens. Matter* **22**, 035404 (2010).
69. Y. Wu, Elasticity of CaSiO_3 - CaTiO_3 perovskite at lower-mantle pressures. *Eur. J. Mineral.* **37**, 773–781 (2025).
70. T. Irifune, A. E. Ringwood, W. O. Hiberson, Subduction of continental crust and terrigenous and pelagic sediments: An experimental study. *Earth Planet. Sci. Lett.* **126**, 351–368 (1994).
71. S. Ono, E. Ito, T. Katsura, Mineralogy of subducted basaltic crust (MORB) from 25 to 37 GPa, and chemical heterogeneity of the lower mantle. *Earth Planet. Sci. Lett.* **190**, 57–63 (2001).
72. L. Stixrude, R. E. Cohen, Y. Rici, H. Krakauer, Prediction of phase transition in CaSiO_3 perovskite and implications for lower mantle structure. *Am. Mineral.* **81**, 1293–1296 (1996).
73. A. M. Dziewonski, D. L. Anderson, Preliminary reference Earth model. *Phys. Earth Planet. In.* **25**, 297–356 (1981).

74. W.-Y. Zhou, M. Hao, W. Su, T. Kim, S. Chen, S.-H. Shim, D. Zhang, P. Q. H. Nguyen, K. Armstrong, J. S. Zhang, Elasticity of davemaoite as a primary contributor to lower-mantle heterogeneities. *Science* **390**, 935–939 (2025).
75. A. K. McNamara, S. Zhong, Thermochemical structures beneath Africa and the Pacific Ocean. *Nature* **437**, 1136–1139 (2005).
76. J. P. Brandenburg, P. E. van Keken, Deep storage of oceanic crust in a vigorously convecting mantle. *J. Geophys. Res. Solid Earth* **112**, B06403 (2007).
77. L. Wang, N. Miyajima, F. Wang, T. Katsura, Persistence of davemaoite at lower-mantle conditions. *Nat. Geosci.* **18**, 365–369 (2025).
78. G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **54**, 11169–11186 (1996).
79. Z. Zhang, R. M. Wentzcovitch, Ab initio anharmonic thermodynamic properties of cubic CaSiO_3 perovskite. *Phys. Rev. B* **103**, 104108 (2021).
80. L. Zhang, J. Han, H. Wang, W. A. Saidi, R. Car, E. Weinan, End-to-end symmetry preserving inter-atomic potential energy model for finite and extended systems. *Adv. Neural Inf. Process. Syst.* **2018**, 4436–4446 (2018).
81. L. Zhang, J. Han, H. Wang, R. Car, E. Weinan, Deep potential molecular dynamics: A scalable model with the accuracy of quantum mechanics. *Phys. Rev. Lett.* **120**, 143001 (2018).
82. A. P. Thompson, H. M. Aktulga, R. Berger, D. S. Bolintineanu, W. M. Brown, P. S. Crozier, P. J. in 't Veld, A. Kohlmeyer, S. G. Moore, T. D. Nguyen, R. Shan, M. J. Stevens, J. Tranchida, C. Trott, S. J. Plimpton, LAMMPS—A flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Comput. Phys. Commun.* **271**, 108171 (2022).
83. W. G. Hoover, B. L. Holian, Kinetic moments method for the canonical ensemble distribution. *Phys. Lett. A* **211**, 253–257 (1996).
84. R. M. Wentzcovitch, Invariant molecular-dynamics approach to structural phase transitions. *Phys. Rev. B* **44**, 2358–2361 (1991).
85. J. R. Ray, A. Rahman, Statistical ensembles and molecular dynamics studies of anisotropic solids. *J. Chem. Phys.* **80**, 4423–4428 (1984).
86. Y. Zhen, C. Chu, A deformation-fluctuation hybrid method for fast evaluation of elastic constants with many-body potentials. *Comput. Phys. Commun.* **183**, 261–265 (2012).
87. G. Clavier, A. P. Thompson, Computation of the thermal elastic constants for arbitrary manybody potentials in LAMMPS using the stress-fluctuation formalism. *Comput. Phys. Commun.* **286**, 108674–108674 (2023).
88. T. H. K. Barron, M. L. Klein, Second-order elastic constants of a solid under stress. *Proc. Phys. Soc.* **85**, 523–532 (1965).
89. C. Luo, J. Tromp, R. M. Wentzcovitch, Ab initio calculations of third-order elastic coefficients. *Phys. Rev. B* **106**, 214104 (2022).
90. S. Cottaar, T. Heister, I. Rose, C. Unterborn, BurnMan: A lower mantle mineral physics toolkit. *Geochem. Geophys. Geosyst.* **15**, 1164–1179 (2014).
91. J. Zeng, D. Zhang, D. Lu, P. Mo, Z. Li, Y. Chen, M. Rynik, L. A. Huang, Z. Li, S. Shi, Y. Wang, H. Ye, P. Tuo, J. Yang, Y. Ding, Y. Li, D. Tisi, Q. Zeng, H. Bao, Y. Xia, J. Huang, K. Muraoka, Y. Wang, J. Chang, F. Yuan, S. L. Bore, C. Cai, Y. Lin, B. Wang, J. Xu, J.-X. Zhu, C. Luo, Y. Zhang, R. E. A. Goodall, W. Liang, A. K. Singh, S. Yao, J. Zhang, R. Wentzcovitch, J. Han, J. Liu, W. Jia, D. M. York, W. E. R. Car, L. Zhang, H. Wang, DeePMD-kit v2: A software package for deep potential models. *J. Chem. Phys.* **159**, 054801 (2023).
92. D. P. Kingma, J. Ba, Adam: A method for stochastic optimization. arXiv:1412.6980 [cs.LG] (2014).
93. H. Wang, L. Zhang, J. Han, W. E. R. Car, DeePMD-kit: A deep learning package for many-body potential energy representation and molecular dynamics. *Comput. Phys. Commun.* **228**, 178–184 (2018).
94. Y. Zhang, H. Wang, W. Chen, J. Zeng, L. Zhang, H. Wang, W. E. R. Car, DP-GEN: A concurrent learning platform for the generation of reliable deep learning based potential energy models. *Comput. Phys. Commun.* **253**, 107206 (2020).
95. J. Liu, G. Zhang, J. Wang, H. Zhang, Y. Han, Research on Cu-Sn machine learning interatomic potential with active learning strategy. *Comput. Mater. Sci.* **246**, 113450 (2025).
96. K. Kawai, T. Tsuchiya, P-V-T equation of state of cubic CaSiO_3 perovskite from first-principles computation. *J. Geophys. Res. Solid Earth* **119**, 2801–2809 (2014).
97. L. Stixrude, C. Lithgow-Bertelloni, Thermodynamics of mantle minerals – I. Physical properties. *Geophys. J. Int.* **162**, 610–632 (2005).
98. Y. Zhang, D. Zhao, M. Matsui, G. Guo, Equations of state of CaSiO_3 perovskite: A molecular dynamics study. *Phys. Chem. Miner.* **33**, 126–137 (2006).
99. M. Noguchi, T. Komabayashi, K. Hirose, Y. Ohishi, High-temperature compression experiments of CaSiO_3 perovskite to lowermost mantle conditions and its thermal equation of state. *Phys. Chem. Miner.* **40**, 81–91 (2013).
100. Y. Kudo, K. Hirose, M. Murakami, Y. Asahara, H. Ozawa, Y. Ohishi, N. Hirao, Sound velocity measurements of CaSiO_3 perovskite to 133 GPa and implications for lowermost mantle seismic anomalies. *Earth Planet. Sci. Lett.* **349–350**, 1–7 (2012).
101. N. Sun, H. Bian, Y. Zhang, J. F. Lin, V. B. Prakapenka, Z. Mao, High-pressure experimental study of tetragonal CaSiO_3 -perovskite to 200 GPa. *Am. Mineral.* **107**, 110–115 (2022).

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Supplementary Materials for
**Ferroelastic hysteresis, shear modulus softening, and the tetragonal↔cubic
transition in davemaoite**

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Supplementary Text S1 and S2
Figs. S1 to S11
Tables S1 and S2
References

Supplementary Text

Text S1: Development of Machine-Learning Potential

The Neural Network potential for Davemaoite (Dm), i.e., cubic CaSiO_3 -perovskite (CaPv), was developed based on the Deep Potential Smooth Edition (DEPPOT-SE) model (80) implemented in DEEPM-D-KIT v2.1 (91). Two-body embedding with coordinates of the neighboring atoms (se_e2_a) was used for the descriptor. The embedding network was designed with a shape of (25, 50, 100), while the fitting network had a shape of (240, 240, 240). The cutoff radius is 6 Å, and the smoothing parameter of 0.5 Å. The model was trained using the Adam optimizer (92) for 1×10^6 training steps, with the learning rate exponentially decaying from 1×10^{-3} to 3.51×10^{-8} during the training process. The loss function $\mathcal{L}(p_e, p_f)$ is given by (93):

$$\mathcal{L}(p_e, p_f) = p_e |\Delta e|^2 + \frac{p_f}{3N} |\Delta f_i|^2,$$

where p_e linearly decays from 1.00 to 0.02, while p_f linearly increases from 1×10^0 to 1×10^3 throughout the training process.

We employed the DP-GEN concurrent learning scheme to create the reference dataset and generate the potential (94). Initially, we randomly extracted 200 labeled configurations from 50 MD runs spanning various P - T ranges, covering 0 – 160 GPa and 300 – 4,000 K to generate the initial potentials. We performed four DP-GEN iterations to explore the configuration space and ultimately achieve a potential that meets the desired accuracy for MD simulations. We trained four candidate DP potentials initialized with different random seeds in each iteration. The error estimator (model deviation) ϵ_t is determined based on the force disagreement between the candidate DPs (94, 95). The expression for ϵ_t is:

$$\epsilon_t = \max_i \sqrt{\langle \|F_{\omega,i}(\mathcal{R}_t) - \langle F_{\omega,i}(\mathcal{R}_t) \rangle\|^2 \rangle},$$

where $F_{\omega,i}(\mathcal{R}_t)$ represents the force on the i -th atom predicted by the ω -th potential for the configuration $\mathcal{R}_t$. For a particular configuration, if ϵ_t satisfies $\epsilon_{\min} \leq \epsilon_t \leq \epsilon_{\max}$, the corresponding configuration is collected, then labeled with DFT forces and total energy then added into the training data set; if $\epsilon_t < \epsilon_{\min}$, these configurations are considered covered by the current training data set; if $\epsilon_t > \epsilon_{\max}$ are considered failed discarded. After a few iterations, almost no new configurations are collected according to this standard (>99% are accurate for a few iterations) and the DP-GEN process is then complete. After these DP-GEN iterations, the final training dataset comprises more than 3,000 configurations annotated with *ab initio* force and energy information. Our DP-SCAN reaches an accuracy of ~ 3.6 meV/atom root mean squared error (RMSE) in energy and ~ 71 meV/Å RMSE in force compared to DFT-SCAN (Fig. S1).

Text S2: Ferroelasticity modelling.

Fig. S11 illustrates the applied stress (σ_c) and lattice parameters ($\mathbf{a}, \mathbf{b}, \mathbf{c}$) during molecular dynamics (MD) simulations at 150 K and 12 GPa, highlighting a structural transition between a lying-down form (LF) and standing form (SF), as indicated by the background colors. The upper panel shows the normal stress along the $\mathbf{c}$ -axis (σ_c), which corresponds to the long side of the tetragonal phase, measured in GPa and plotted as a function of MD simulation time (ps). Initially, an external compressive stress was applied to the system, increasing gradually from 0 GPa to 5 GPa. After reaching this peak, the stress decreases until it reaches zero and further decreases, becoming tensile until a maximum tensile stress of 5 GPa is achieved. Finally, the applied tensile stress is gradually reduced until the system returns to its initial stress-free state. The switching between the two orientation states of CaPv, induced by the external stress σ_c , occurs at specific values of compressive and tensile stress. The lower panel shows the time-dependent lattice parameters ($\mathbf{a}, \mathbf{b}, \mathbf{c}$) in angstroms. Notably, upon entering the lying-down form, the $\mathbf{a}$ and $\mathbf{b}$ parameters show significant divergence, with $\mathbf{b} \neq \mathbf{c}$. This occurs because the system transitions from a state where the $\mathbf{c}$ -axis is the long axis of the tetragonal phase ($\mathbf{a} = \mathbf{b} < \mathbf{c}$) to one where the $\mathbf{a}$ -axis becomes the long axis ($\mathbf{b} = \mathbf{c} < \mathbf{a}$), caused by the applied compressive stress along the $\mathbf{c}$ -axis. The purple dashed line in the figure indicates that when σ_c returns to zero, the system reverts to the $\mathbf{b} = \mathbf{c} < \mathbf{a}$ tetragonal phase. This suggests that the transition is characterized by a switch in the long axis of the tetragonal phase between the $\mathbf{a}$ - and $\mathbf{c}$ -axes.

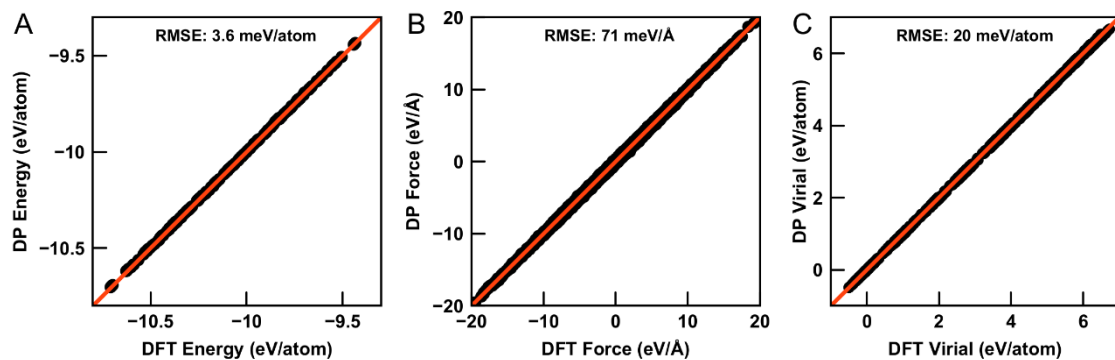


Fig. S1. Validation and accuracy assessment of the DP-SCAN. Comparison of (A) potential energies, (B) atomic forces, and (C) virial from DP and DFT calculations.

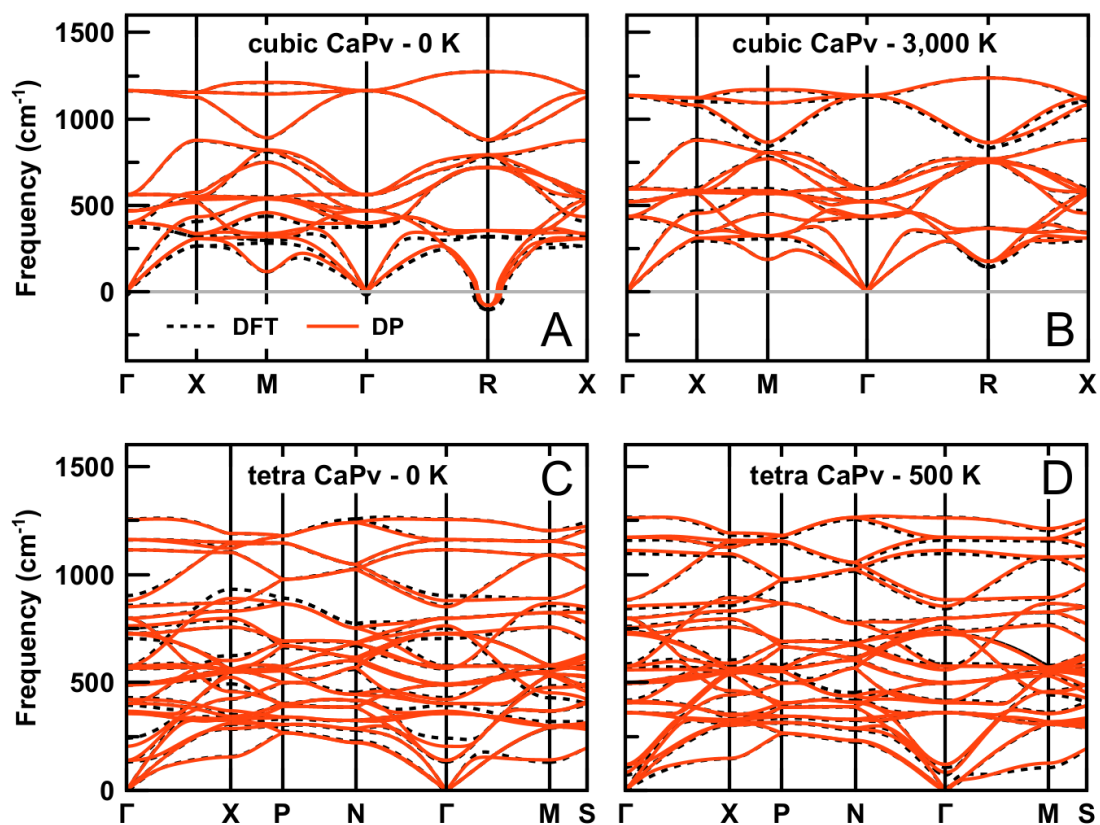


Fig. S2. Phonon dispersion relations of cubic and tetragonal CaPv. (A) Harmonic phonon dispersions at 95 GPa and 0 K. (B) anharmonic phonon dispersions at 95 GPa and 3,000 K obtained using the DP-potential and SCAN-DFT for cubic CaPv. (C) Harmonic phonon dispersions at 95 GPa and 0 K. (D) anharmonic phonon dispersions at 95 GPa and 500 K by DP and DFT for tetragonal CaPv.

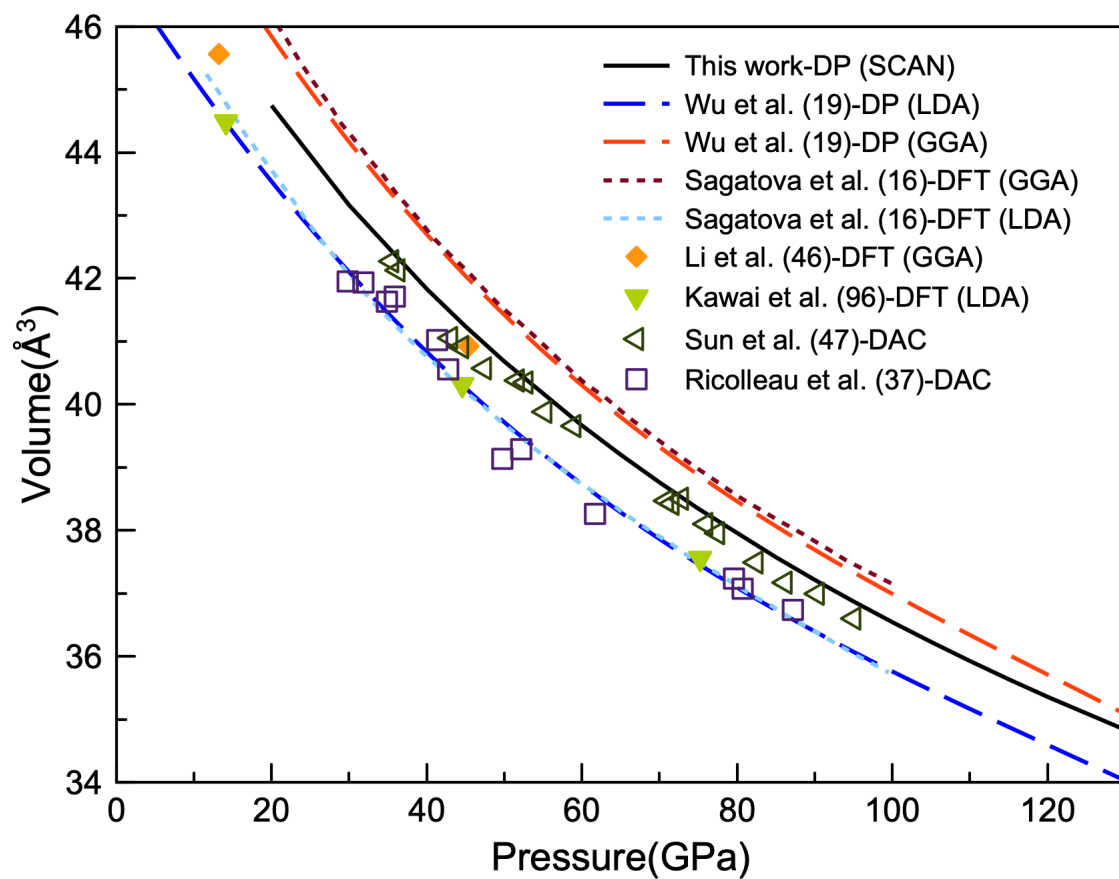


Fig. S3. Compression curve of cubic CaPv at 2,000 K. Solid curves are obtained using DP-SCAN in this work, and dashed lines are the previous calculations (16, 19, 46, 96). Previous measurements (37, 47) are also plotted for comparison.

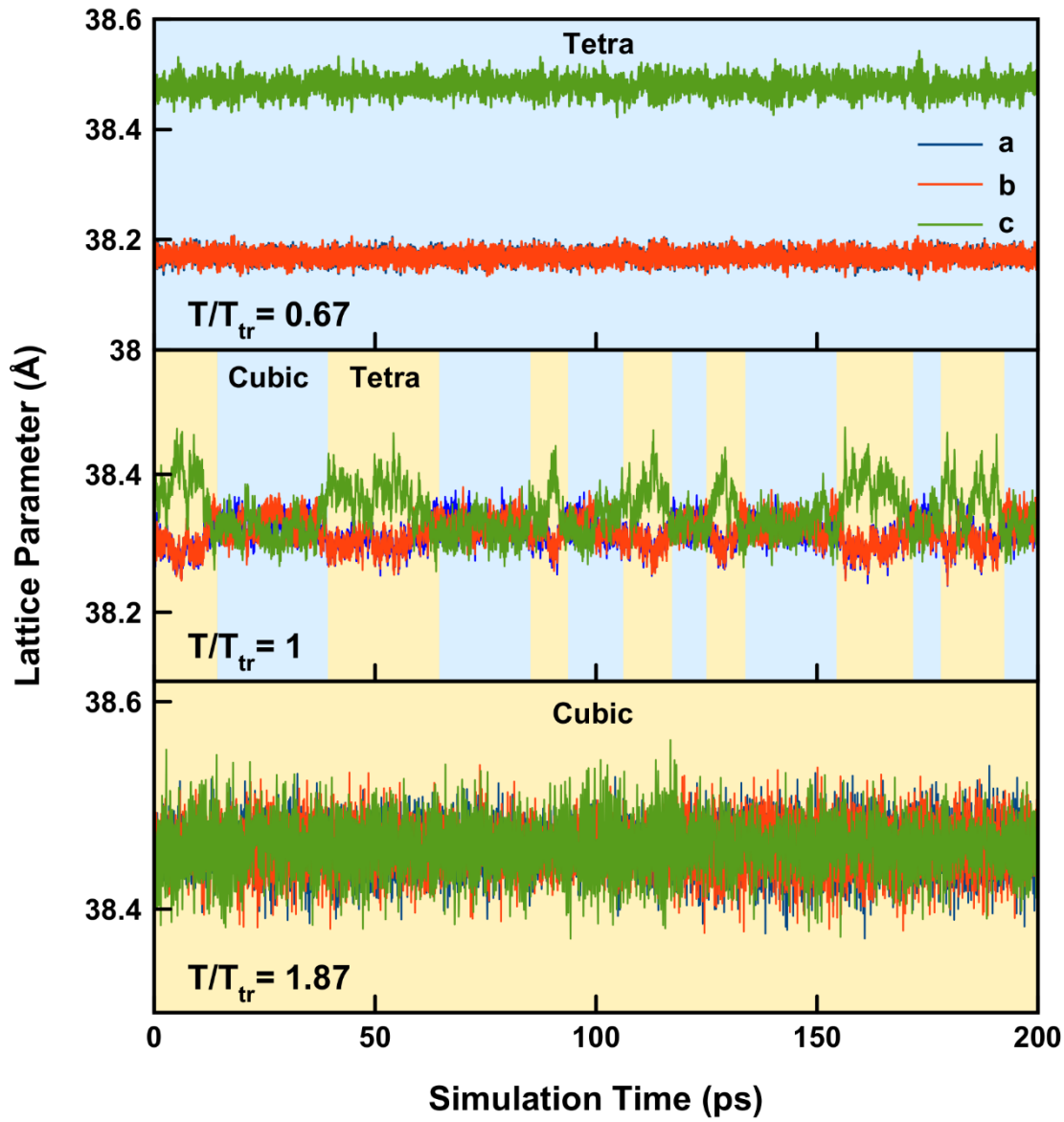


Fig. S4. Fluctuations of lattice parameters (a , b , c) during MD simulations at 60 GPa. The temperatures correspond to $0.67 \times T_{tr}$, $1 \times T_{tr}$, and $1.87 \times T_{tr}$. Near the transition temperature, the system frequently fluctuates between the tetragonal phase (yellow region) and the cubic phase (blue region). At temperatures well below T_{tr} , the system remains in the tetragonal phase, whereas at temperatures far above T_{tr} , it stabilizes in the cubic phase.

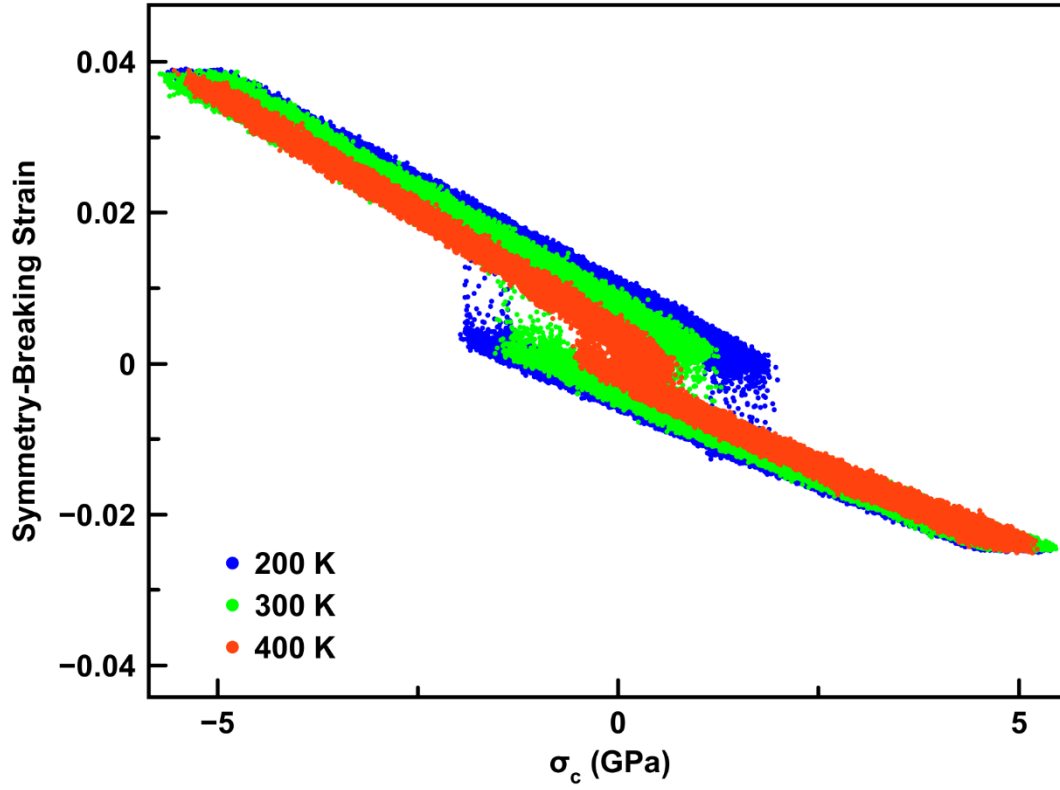


Fig. S5. Temperature effect on the ferroelastic behavior of CaPv. The applied stress (σ_c) and symmetry-breaking strain $e_t = [(c - a)/a]$ during molecular dynamics (MD) simulations were obtained at 12 GPa and 1 different temperatures. Symbol colors denote temperatures.

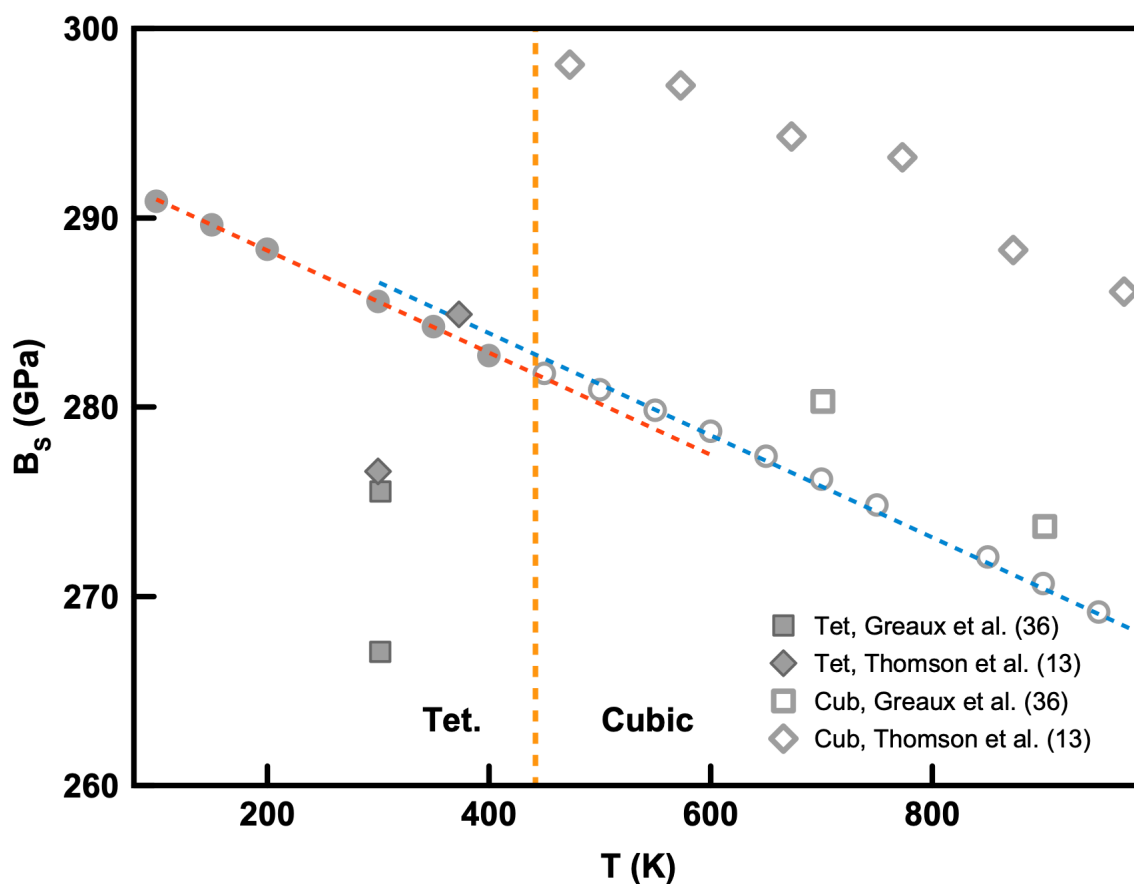


Fig. S6. Bulk modulus (B_s) vs temperature at 12 GPa. Filled symbols denote the tetragonal CaPv, while open symbols denote the cubic CaPv. Red and blue dashed lines are fitted linear curve for tetragonal and cubic CaPv, respectively. Measured elastic moduli from previous studies are plotted as grey diamonds (13) (about 12 GPa) and grey squares (36) (12 ± 1 GPa). Orange dashed lines indicates the phase transition line measured by Thomson et al. (13).

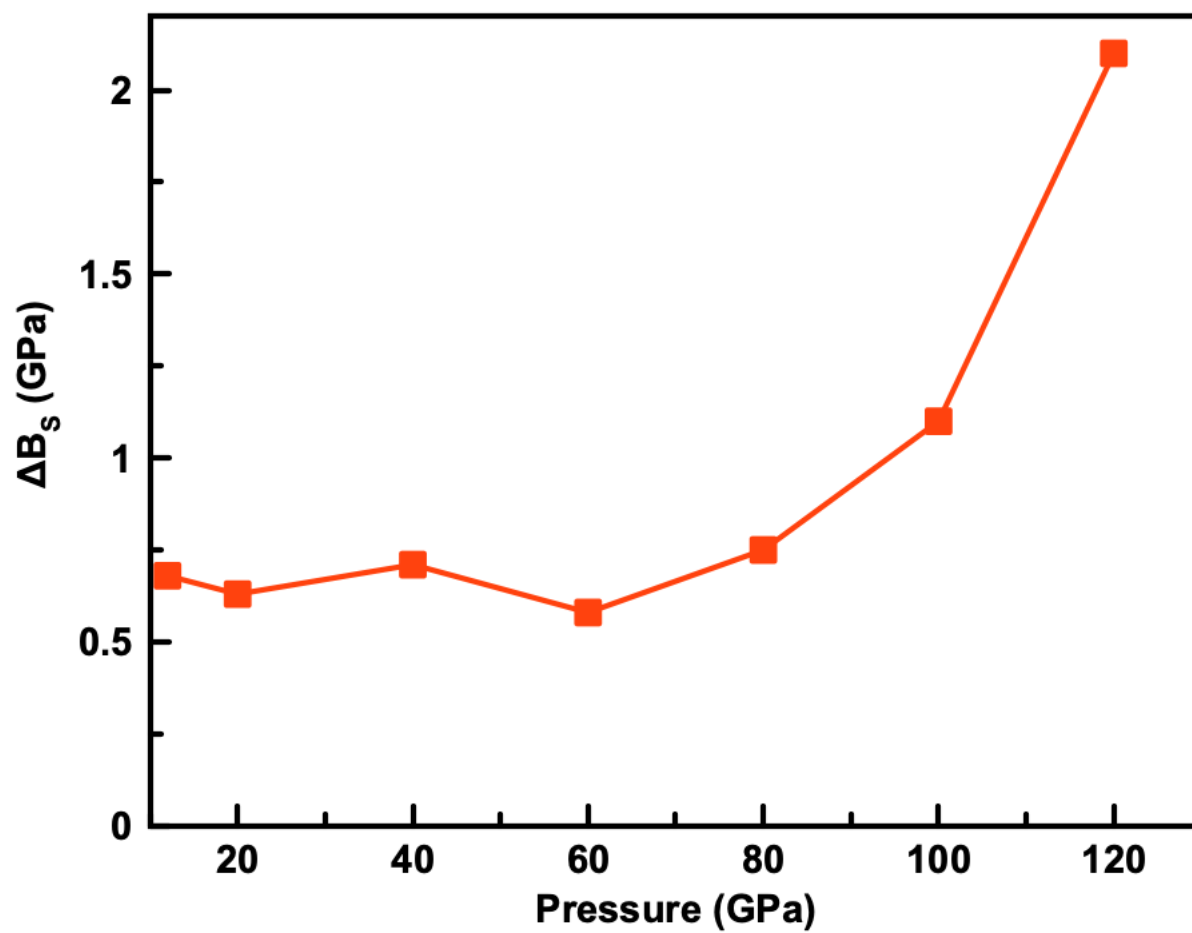
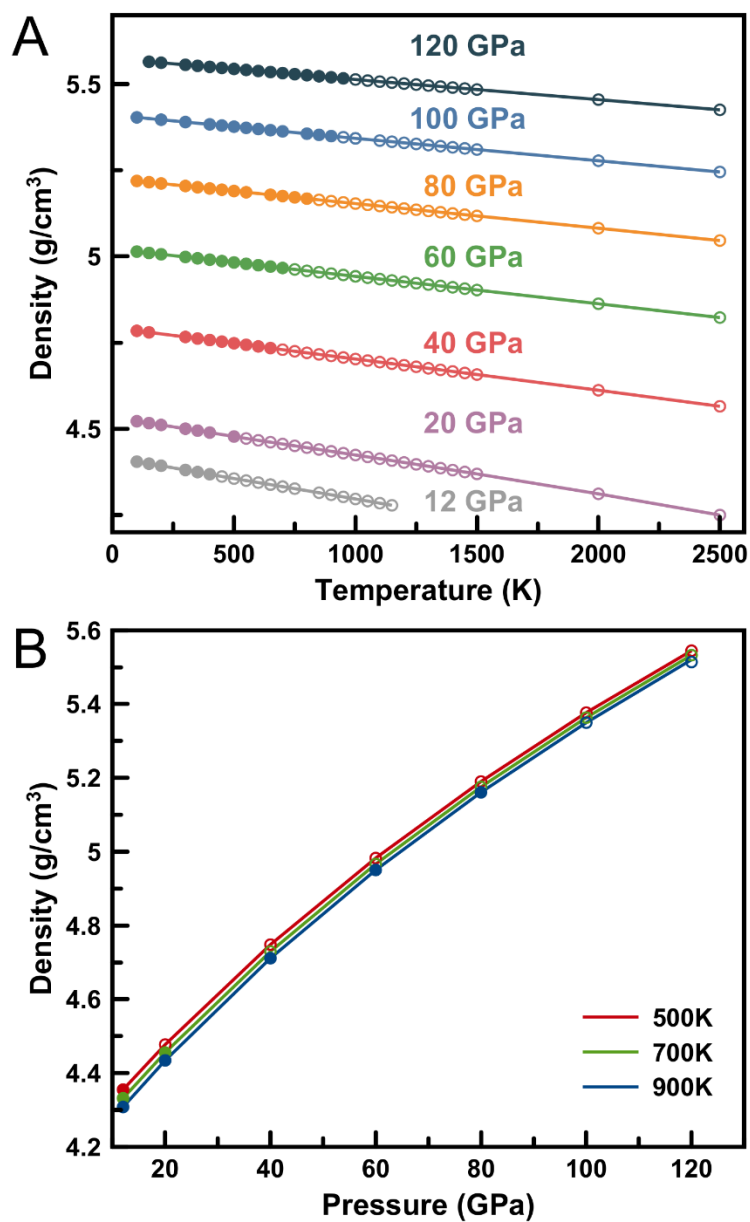
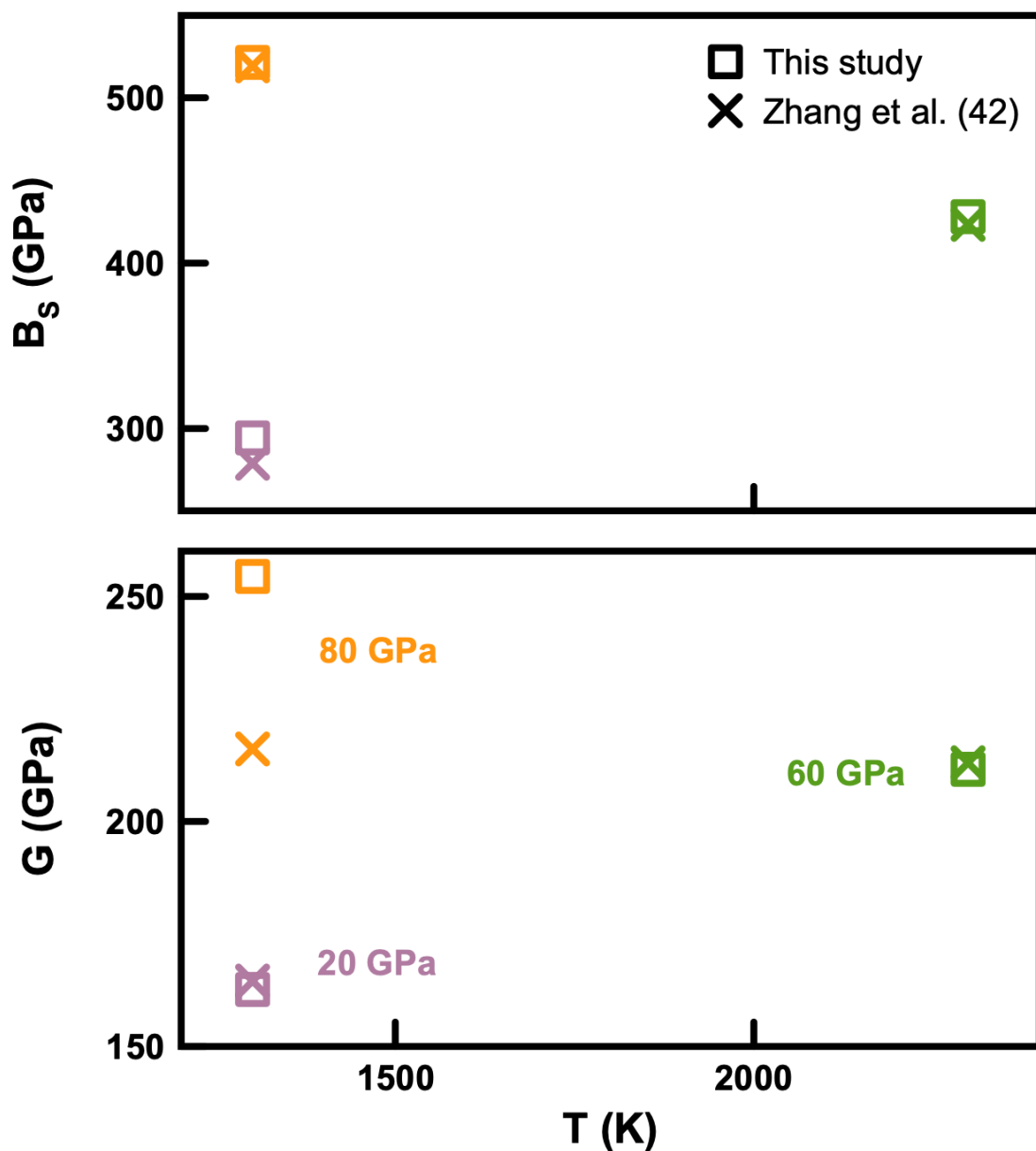


Fig. S7. Reduction in the adiabatic bulk modulus (ΔB_s) near the tetragonal $\leftrightarrow$ cubic phase transition.



1 **Fig. S8. Density of CaPv across the ferroelastic↔paraelastic transition.** (A) Density vs Temperature. Colors
2 denote different pressures. (B) Density vs Pressure. Colors denote different temperatures. Filled symbols denote
3 the tetragonal CaPv, while open symbols denote the cubic CaPv.



5 **Fig. S9. Comparison of bulk modulus and shear modulus vs temperature.** Colors denote different pressures.
6 Crosses denote the results from Zhang et al. (42), while squares denote the results from this study. The difference
7 at 80 GPa results from the softening effect near the transition temperature, which is higher in Zhang et al. 's result.

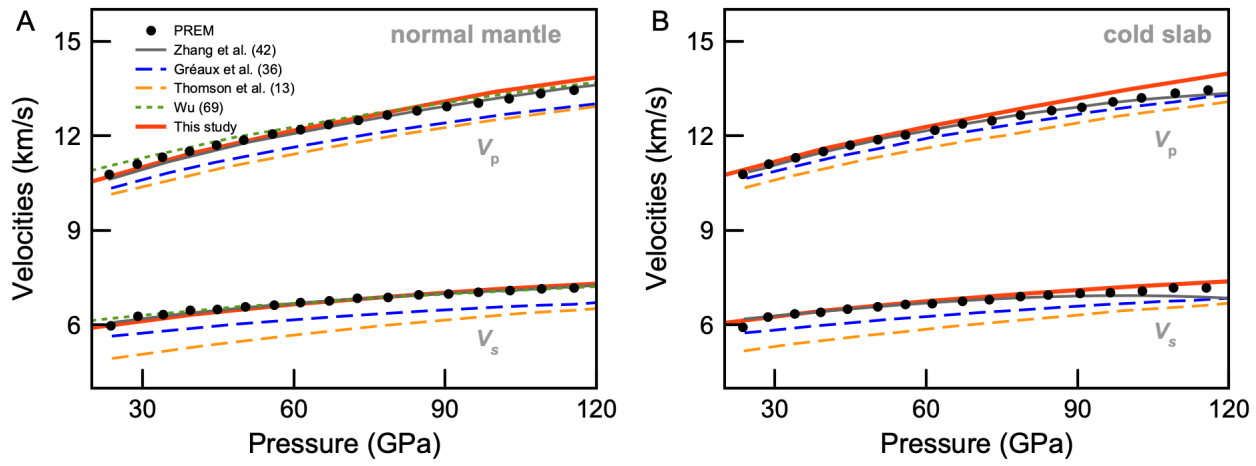


Fig. S10. Seismic wave velocities of CaPv along normal mantle and cold subducting slab geotherms. (A) Velocity profiles along the geotherms of 1,600 K adiabat representing the normal mantle. The green dotted line represents the wave velocities of $\text{Ca}(\text{Si,Ti})\text{O}_3$ solid solutions from Wu's static first-principles calculations at 0 K (69). (B) Velocity profiles along a 1,200 K adiabat representing the subducted cold slab (90).

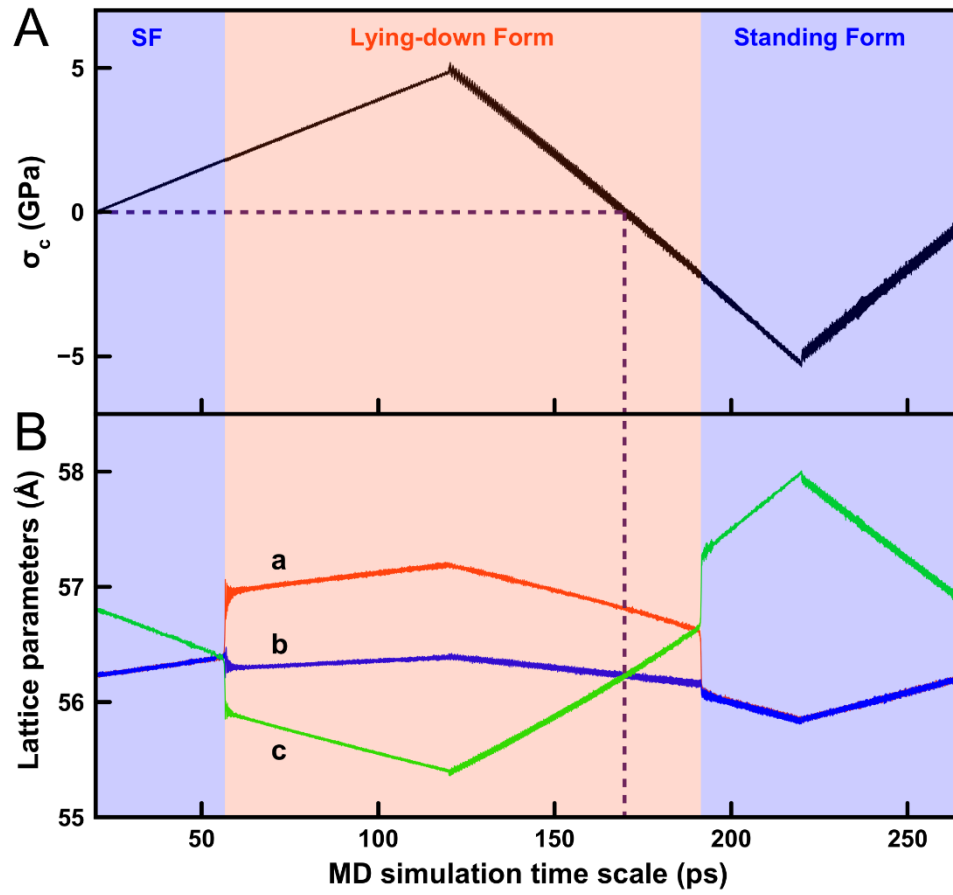


Fig. S11. Evolution of stress (σ_c) and lattice parameters (a, b, c), where (a, b, c) represents the lattice vectors of the tetragonal primitive cell ($\sqrt{2} \times \sqrt{2} \times 2$, 20 atoms), during MD simulations at 12 GPa and 150 K. (A) The normal stress in GPa along the c-axis (σ_c) of the SF state was plotted as a function of MD simulation time. Purple dashed line indicates the stress-free state ($\sigma_c=0$); (B) The external stress (σ_c) dependence of the lattice parameters (a, b, c) in Angstroms.

Table S1. Thermal EoS parameters of CaPv.

Ref.	Phase	T_{ref} (K)	V_0 (Å ³)	K_{T0} (GPa)	K'	θ_0 (K)	γ_0	q	G_0 (GPa)	G'	η_{s0}
<i>Theoretical calculations</i>											
This study*	Tet.	300	46.2	226	4.3	292	1.9	1.86	133.5	1.6	11.9
This study*	Cub.	1000	47.4	200	4.46	694	1.63	1.3	131.6	1.99	1.65
Zhang et al., 2006 ^a	Cub.	300	45.6	242	4.18	1100 [#]	1.52	0.97	-	-	-
Li et al., 2006 ^b	Cub.	2000	48.7	171	4.76	-	1.63	1	-	-	-
Tsuchiya et al., 2014 ^c	Cub.	1000	46.2	207	4.41	1100 [#]	1.57	0.84	-	-	-
Zhang et al., 2025 ^d	Cub.	700	46.2	208	-	1000 [#]	1.65	1.03	-	-	-
<i>Experiments</i>											
Noguchi et al., 2013 ^e	Cub.	700	56.5	238	4 [#]	1300	2.8	1.2	-	-	-
Sun et al., 2016 ^f	Cub.	300	45.4	249	4 [#]	1000 [#]	1.8	1.1	-	-	-
Gréaux et al., 2019 ^g	Cub.	700	45.6 [#]	258	4 [#]	796 [#]	1.51	6.5	125	1.62	1.02
Thomson et al., 2019 ^h	Cub.	300	45.6 [#]	258	3.63	778	1.67	1.09	111.3	1.66 [#]	4.8

* Simulation results were fitted to the Mie-Grüneisen-Debye equation of state (97).

[#] Fixed parameter.

^a Ref. [(98)]. Classical Molecular Dynamics (MD) calculations

^b Ref. [(17, 46)]. First-Principle Molecular Dynamics (FPMD) calculations with GGA.

^c Ref. [(96)]. FPMD calculations with LDA.

^d Ref. [(42)]. Machine-Learning Force Fields (MLFF) Molecular Dynamics(MD) calculations with LDA with a pressure correction of 4.3 GPa.

^e Ref. [(99)].

^f Ref. [(47)].

^g Ref. [(36)]. Data refitted to extract EoS parameters.

^h Ref. [(13)].

Table S2. Isothermal EoS parameters of tetragonal CaPv.

Ref.	T (K)	V_0 (Å ³)	K_{T0} (GPa)	K'
<i>Theoretical calculations</i>				
This study*	300	46.2	227	4.2
Caracas et al., 2005 ^a	0	44.5	249	4.09
Tsuchiya et al., 2011 ^b	0	44.6	244	4.44
Wu et al., 2024 ^c	300	44.6	255	3.97
Wu et al., 2024 ^d	300	47.0	238	3.69
Zhang et al., 2025 ^e	300	45.6	223	4.23
<i>Experiments</i>				
Kudo et al., 2012 ^f	300	45.6	244	4
Chen et al., 2018 ^g	300	46.3	223	4
Gréaux et al., 2019 ^h	300	45.6	228	3.7
Thomson et al., 2019 ⁱ	300	46.1	224	4
Sun et al., 2022 ^j	300	45.6	227	4

* Data and results were fitted to a 3rd order Birch-Murnaghan 3rd equation of state

^a Ref. [(18)]. First-Principle static calculations with LDA.

^b Ref. [(44)]. First-Principle static calculations with LDA.

^c Ref. [(19)]. Deep Potential Molecular Dynamics with LDA.

^d Ref. [(19)]. Deep Potential Molecular Dynamics with GGA.

^e Ref. [(42)]. MLFF MD calculations with LDA with a pressure correction of 4.3 GPa.

^f Ref. [(100)]

^g Ref. [(35)]

^h Ref. [(36)]

ⁱ Ref. [(13)]

^j Ref. [(101)]

REFERENCES

1. T. Irifune, Absence of an aluminous phase in the upper part of the Earth's lower mantle. *Nature* **370**, 131–133 (1994).
2. A. Saikia, D. J. Frost, D. C. Rubie, Splitting of the 520-kilometer seismic discontinuity and chemical heterogeneity in the mantle. *Science* **319**, 1515–1518 (2008).
3. K. Hirose, Y. Fei, Subsolidus and melting phase relations of basaltic composition in the uppermost lower mantle. *Geochim. Cosmochim. Acta* **66**, 2099–2108 (2002).
4. T. Irifune, A. E. Ringwood, Phase transformations in subducted oceanic crust and buoyancy relationships at depths of 600–800 km in the mantle. *Earth Planet. Sci. Lett.* **117**, 101–110 (1993).
5. S. E. Kesson, J. D. Fitz Gerald, J. M. G. Shelley, Mineral chemistry and density of subducted basaltic crust at lower-mantle pressures. *Nature* **372**, 767–769 (1994).
6. T. Irifune, M. Miyashita, T. Inoue, J. Ando, K. Funakoshi, W. Utsumi, High-pressure phase transformation in $\text{CaMgSi}_2\text{O}_6$ and implications for origin of ultra-deep diamond inclusions. *Geophys. Res. Lett.* **27**, 3541–3544 (2000).
7. L. Vitos, B. Magyari-Köpe, R. Ahuja, J. Kollár, G. Grimvall, B. Johansson, Phase transformations between garnet and perovskite phases in the Earth's mantle: A theoretical study. *Phys. Earth Planet. In.* **156**, 108–116 (2006).
8. D. Y. Jung, M. W. Schmidt, Solid solution behaviour of CaSiO_3 and MgSiO_3 perovskites. *Phys. Chem. Miner.* **38**, 311–319 (2011).
9. J. M. R. Muir, A. R. Thomson, F. Zhang, The miscibility of calcium silicate perovskite and bridgmanite: A single perovskite solid solution in hot, iron-rich regions. *Earth Planet. Sci. Lett.* **566**, 116973–116973 (2021).
10. B. Ko, E. Greenberg, V. Prakapenka, E. E. Alp, W. Bi, Y. Meng, D. Zhang, S.-H. Shim, Calcium dissolution in bridgmanite in the Earth's deep mantle. *Nature* **611**, 88–92 (2022).

11. K.-H. Chao, M. Berrada, S. Wang, J. Peckenpaugh, D. Zhang, S. Chariton, V. Prakapenka, B. Chen, Structure and equation of state of Ti-bearing davemaoite: New insights into the chemical heterogeneity in the lower mantle. *Am. Mineral.* **109**, 1861–1870 (2024).
12. A. R. Thomson, M. J. Walter, S. C. Kohn, R. A. Brooker, Slab melting as a barrier to deep carbon subduction. *Nature* **529**, 76–79 (2016).
13. A. R. Thomson, W. A. Crichton, J. P. Brodholt, I. G. Wood, N. C. Siersch, J. M. R. Muir, D. P. Dobson, S. A. Hunt, Seismic velocities of CaSiO₃ perovskite can explain LLSVPs in Earth's lower mantle. *Nature* **572**, 643–647 (2019).
14. T. Komabayashi, K. Hirose, N. Sata, Y. Ohishi, L. S. Dubrovinsky, Phase transition in CaSiO₃ perovskite. *Earth Planet. Sci. Lett.* **260**, 564–569 (2007).
15. T. Kurashina, K. Hirose, S. Ono, N. Sata, Y. Ohishi, Phase transition in Al-bearing CaSiO₃ perovskite: Implications for seismic discontinuities in the lower mantle. *Phys. Earth Planet. In.* **145**, 67–74 (2004).
16. D. N. Sagatova, A. F. Shatskiy, N. E. Sagatov, K. D. Litasov, Phase Relations in CaSiO₃ System up to 100 GPa and 2500 K. *Geochem. Int.* **59**, 791–800 (2021).
17. L. Li, D. J. Weidner, J. Brodholt, D. Alfè, G. David Price, R. Caracas, R. Wentzcovitch, Elasticity of CaSiO₃ perovskite at high pressure and high temperature. *Phys. Earth Planet. In.* **155**, 249–259 (2006).
18. R. Caracas, R. Wentzcovitch, G. D. Price, J. Brodholt, CaSiO₃ perovskite at lower mantle pressures. *Geophys. Res. Lett.* **32**, 1–5 (2005).
19. F. Wu, Y. Sun, T. Wan, S. Wu, R. M. Wentzcovitch, Deep-learning-based prediction of the tetragonal → cubic transition in davemaoite. *Geophys. Res. Lett.* **51**, e2023GL108012 (2024).
20. M. A. Carpenter, E. K. H. Salje, Elastic anomalies in minerals due to structural phase transitions. *Eur. J. Mineral.* **10**, 693–812 (1998).

21. E. Salje, G. Hoppmann, Direct observation of ferroelasticity in $\text{Pb}_3(\text{PO}_4)_2 \cdot \text{Pb}_3(\text{VO}_4)_2$. *Mater. Res. Bull.* **11**, 1545–1549 (1976).
22. Y. Zhang, S. Fu, B. Wang, J. F. Lin, Elasticity of a pseudoproper ferroelastic transition from stishovite to post-stishovite at high pressure. *Phys. Rev. Lett.* **126**, 025701–025701 (2021).
23. M. A. Carpenter, R. J. Hemley, H. K. Mao, High-pressure elasticity of stishovite and the $\text{P42/mmm} \rightarrow \text{Pnnm}$ phase transition. *J. Geophys. Res. Solid Earth* **105**, 10807–10816 (2000).
24. L. G. Liu, A. E. Ringwood, Synthesis of a perovskite-type polymorph of CaSiO_3 . *Earth Planet. Sci. Lett.* **28**, 209–211 (1975).
25. J. Deng, L. Stixrude, Thermal conductivity of silicate liquid determined by machine learning potentials. *Geophys. Res. Lett.* **48**, e2021GL093806 (2021).
26. Z. Zhang, G. Csányi, D. Alfè, Partitioning of sulfur between solid and liquid iron under Earth's core conditions: Constraints from atomistic simulations with machine learning potentials. *Geochim. Cosmochim. Acta* **291**, 5–18 (2020).
27. F. Xu, D. Wang, Z. Li, H. Song, L. Liu, H. Geng, J. Hu, X. Chen, Large-scale simulation of thermal conductivity in CaSiO_3 perovskite with neuroevolution potential. *Appl. Phys. Lett.* **125**, 5–5 (2024).
28. Z. Zhang, D. B. Zhang, T. Sun, R. M. Wentzcovitch, phq: A Fortran code to compute phonon quasiparticle properties and dispersions. *Comput. Phys. Commun.* **243**, 110–120 (2019).
29. D. B. Zhang, T. Sun, R. M. Wentzcovitch, Phonon quasiparticles and anharmonic free energy in complex systems. *Phys. Rev. Lett.* **112**, 058501 (2014).
30. J. Sun, A. Ruzsinszky, J. P. Perdew, Strongly constrained and appropriately normed semilocal density functional. *Phys. Rev. Lett.* **115**, 036402 (2015).
31. C. Luo, Y. Sun, R. Wentzcovitch, Elasticity and acoustic velocities of $\delta\text{-AlOOH}$ at extreme conditions: A methodology assessment. *Phys. Rev. Mater.* **8**, 103601 (2024).

32. H. K. Mao, L. C. Chen, R. J. Hemley, A. P. Jephcoat, Y. Wu, W. A. Bassett, Stability and equation of state of CaSiO_3 -perovskite to 134 GPa. *J. Geophys. Res.* **94**, 17889–17894 (1989).
33. Y. Wang, D. J. Weidner, F. Guyot, Thermal equation of state of CaSiO_3 perovskite. *J. Geophys. Res. Solid Earth* **101**, 661–672 (1996).
34. S. H. Shim, R. Jeanloz, T. S. Duffy, Tetragonal structure of CaSiO_3 perovskite above 20 GPa. *Geophys. Res. Lett.* **29**, 2166–2166 (2002).
35. H. Chen, S. H. Shim, K. Leinenweber, V. Prakapenka, Y. Meng, C. Prescher, Crystal structure of CaSiO_3 perovskite at 28–62 GPa and 300 K under quasi-hydrostatic stress conditions. *Am. Mineral.* **103**, 462–468 (2018).
36. S. Gréaux, T. Irifune, Y. Higo, Y. Tange, T. Arimoto, Z. Liu, A. Yamada, Sound velocity of CaSiO_3 perovskite suggests the presence of basaltic crust in the Earth's lower mantle. *Nature* **565**, 218–221 (2019).
37. A. Ricolleau, J. P. Perrillat, G. Fiquet, I. Daniel, J. Matas, A. Addad, N. Menguy, H. Cardon, M. Mezouar, N. Guignot, Phase relations and equation of state of a natural MORB: Implications for the density profile of subducted oceanic crust in the Earth's lower mantle. *J. Geophys. Res. Solid Earth* **115**, B08202 (2010).
38. E. K. H. Salje, Ferroelastic materials. *Annu. Rev. Mat. Res.* **42**, 265–283 (2012).
39. S. Kaneshima, Seismic scatterers in the mid-lower mantle. *Phys. Earth Planet. In.* **257**, 105–114 (2016).
40. E. K. H. Salje, Application of Landau theory for the analysis of phase transitions in minerals. *Phys. Rep.* **215**, 49–99 (1992).
41. J. Zhuang, R. Wentzcovitch, Fe^{2+} partitioning in al-free pyrolite: Consequences for seismic velocities and heterogeneities. *Geophys. Res. Lett.* **51**, e2024GL108967 (2024).

42. C. Zhang, J.-Y. Yang, T. Sun, H. Zhang, J. P. Brodholt, Strong precursor softening in cubic CaSiO_3 perovskite. *Proc. Natl. Acad. Sci. U.S.A.* **122**, e2410910122 (2025).
43. L. Stixrude, C. Lithgow-Bertelloni, B. Kiefer, P. Fumagalli, Phase stability and shear softening in CaSiO_3 perovskite at high pressure. *Phys. Rev. B* **75**, 024108 (2007).
44. T. Tsuchiya, Elasticity of subducted basaltic crust at the lower mantle pressures: Insights on the nature of deep mantle heterogeneity. *Phys. Earth Planet. In.* **188**, 142–149 (2011).
45. M. A. Carpenter, Elastic anomalies accompanying phase transitions in $(\text{Ca,Sr})\text{TiO}_3$ perovskites: Part I. Landau theory and a calibration for SrTiO_3 . *Am. Mineral.* **92**, 309–327 (2007).
46. L. Li, D. J. Weidner, J. Brodholt, D. Alfè, G. D. Price, R. Caracas, R. Wentzcovitch, Phase stability of CaSiO_3 perovskite at high pressure and temperature: Insights from ab initio molecular dynamics. *Phys. Earth Planet. In.* **155**, 260–268 (2006).
47. N. Sun, Z. Mao, S. Yan, X. Wu, V. B. Prakapenka, J. F. Lin, Confirming a pyrolitic lower mantle using self-consistent pressure scales and new constraints on CaSiO_3 perovskite. *J. Geophys. Res. Solid Earth* **121**, 4876–4894 (2016).
48. Y. Shin, E. Di Lucente, N. Marzari, L. Monacelli, The thermodynamics of CaSiO_3 perovskite in Earth's lower mantle. *Phys. Rev. B* **112**, 174113 (2025).
49. N. J. Perks, Z. Zhang, R. J. Harrison, M. A. Carpenter, Strain relaxation mechanisms of elastic softening and twin wall freezing associated with structural phase transitions in $(\text{Ca,Sr})\text{TiO}_3$ perovskites. *J. Phys. Condens. Matter* **26**, 505402 (2014).
50. E. M. Syracuse, P. E. van Keken, G. A. Abers, D. Suetsugu, C. Bina, T. Inoue, D. Wiens, M. Jellinek, The global range of subduction zone thermal models. *Phys. Earth Planet. In.* **183**, 73–90 (2010).
51. T. Wan, C. Luo, Y. Sun, R. M. Wentzcovitch, Thermoelastic properties of bridgmanite using deep-potential molecular dynamics. *Phys. Rev. B* **109**, 094101 (2024).

52. A. Yeganeh-Haeri, Synthesis and re-investigation of the elastic properties of single-crystal magnesium silicate perovskite. *Phys. Earth Planet. In.* **87**, 111–121 (1994).
53. S. V. Sinogeikin, J. Zhang, J. D. Bass, Elasticity of single crystal and polycrystalline MgSiO_3 perovskite by Brillouin spectroscopy. *Geophys. Res. Lett.* **31**, e2004GL019559 (2004).
54. G. Criniti, A. Kurnosov, T. Boffa Ballaran, D. J. Frost, Single-crystal elasticity of MgSiO_3 bridgmanite to mid-lower mantle pressure. *J. Geophys. Res. Solid Earth* **126**, e022020JB020967 (2021).
55. R. M. Wentzcovitch, B. B. Karki, M. Cococcioni, S. de Gironcoli, Thermoelastic properties of MgSiO_3 -perovskite: Insights on the nature of the Earth's lower mantle. *Phys. Rev. Lett.* **92**, 018501 (2004).
56. A. R. Oganov, J. P. Brodholt, G. D. Price, Ab initio elasticity and thermal equation of state of MgSiO_3 perovskite. *Earth Planet. Sci. Lett.* **184**, 555–560 (2001).
57. B. Wehinger, A. Bosak, S. Nazzareni, D. Antonangeli, A. Mirone, S. L. Chaplot, R. Mittal, E. Ohtani, A. Shatskiy, S. Saxena, S. Ghose, M. Krisch, Dynamical and elastic properties of MgSiO_3 perovskite (bridgmanite). *Geophys. Res. Lett.* **43**, 2568–2575 (2016).
58. K. Kawai, T. Tsuchiya, Small shear modulus of cubic CaSiO_3 perovskite. *Geophys. Res. Lett.* **42**, 2718–2726 (2015).
59. A. Nowacki, J. Wookey, J. M. Kendall, Deformation of the lowermost mantle from seismic anisotropy. *Nature* **467**, 1091–1094 (2010).
60. M. D. Long, Complex anisotropy in D'' beneath the eastern Pacific from SKS–SKKS splitting discrepancies. *Earth Planet. Sci. Lett.* **283**, 181–189 (2009).
61. L. Miyagi, W. Kanitpanyacharoen, P. Kaercher, K. K. M. Lee, H.-R. Wenk, Slip systems in MgSiO_3 post-perovskite: Implications for D'' anisotropy. *Science* **329**, 1639–1641 (2010).

62. S. B. Merkel, A. K. McNamara, A. Kubo, S. Speziale, L. Miyagi, Y. Meng, T. S. Duffy, H.-R. Wenk, Deformation of (Mg,Fe)SiO₃ post-perovskite and D'' anisotropy. *Science* **316**, 1729–1732 (2007).
63. A. Kohutych, R. Yevych, S. Perechinskii, V. Samulionis, J. Banys, Y. Vysochanskii, Sound behavior near the Lifshitz point in proper ferroelectrics. *Phys. Rev. B* **82**, 054101 (2010).
64. R. J. Harrison, S. A. T. Redfern, J. Street, The effect of transformation twins on the seismic-frequency mechanical properties of polycrystalline Ca_{1-x}Sr_xTiO₃ perovskite. *Am. Mineral.* **88**, 574–582 (2003).
65. M. A. Carpenter, Elastic properties of minerals and the influence of phase transitions. *Am. Mineral.* **91**, 229–246 (2006).
66. M. A. Carpenter, Z. Zhang, Anelasticity maps for acoustic dissipation associated with phase transitions in minerals. *Geophys. J. Int.* **186**, 279–295 (2011).
67. Z. Zhang, J. Koppensteiner, W. Schranz, D. Prabhakaran, M. A. Carpenter, Strain coupling mechanisms and elastic relaxation associated with spin state transitions in LaCoO₃. *J. Phys. Condens. Matter* **23**, 145401 (2011).
68. M. A. Carpenter, S. V. Sinogeikin, J. D. Bass, Elastic relaxations associated with the Pm $\bar{3}$ m–R $\bar{3}$ c transition in LaAlO₃: II. Mechanisms of static and dynamical softening. *J. Phys. Condens. Matter* **22**, 035404 (2010).
69. Y. Wu, Elasticity of CaSiO₃–CaTiO₃ perovskite at lower-mantle pressures. *Eur. J. Mineral.* **37**, 773–781 (2025).
70. T. Irifune, A. E. Ringwood, W. O. Hibberson, Subduction of continental crust and terrigenous and pelagic sediments: An experimental study. *Earth Planet. Sci. Lett.* **126**, 351–368 (1994).
71. S. Ono, E. Ito, T. Katsura, Mineralogy of subducted basaltic crust (MORB) from 25 to 37 GPa, and chemical heterogeneity of the lower mantle. *Earth Planet. Sci. Lett.* **190**, 57–63 (2001).

72. L. Stixrude, R. E. Cohen, Y. Rici, H. Krakauer, Prediction of phase transition in CaSiO_3 perovskite and implications for lower mantle structure. *Am. Mineral.* **81**, 1293–1296 (1996).
73. A. M. Dziewonski, D. L. Anderson, Preliminary reference Earth model. *Phys. Earth Planet. In.* **25**, 297–356 (1981).
74. W.-Y. Zhou, M. Hao, W. Su, T. Kim, S. Chen, S.-H. Shim, D. Zhang, P. Q. H. Nguyen, K. Armstrong, J. S. Zhang, Elasticity of davemaoite as a primary contributor to lower-mantle heterogeneities. *Science* **390**, 935–939 (2025).
75. A. K. McNamara, S. Zhong, Thermochemical structures beneath Africa and the Pacific Ocean. *Nature* **437**, 1136–1139 (2005).
76. J. P. Brandenburg, P. E. van Keken, Deep storage of oceanic crust in a vigorously convecting mantle. *J. Geophys. Res. Solid Earth* **112**, B06403 (2007).
77. L. Wang, N. Miyajima, F. Wang, T. Katsura, Persistence of davemaoite at lower-mantle conditions. *Nat. Geosci.* **18**, 365–369 (2025).
78. G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **54**, 11169–11186 (1996).
79. Z. Zhang, R. M. Wentzcovitch, Ab initio anharmonic thermodynamic properties of cubic CaSiO_3 perovskite. *Phys. Rev. B* **103**, 104108 (2021).
80. L. Zhang, J. Han, H. Wang, W. A. Saidi, R. Car, E. Weinan, End-to-end symmetry preserving inter-atomic potential energy model for finite and extended systems. *Adv. Neural Inf. Process. Syst.* **2018**, 4436–4446 (2018).
81. L. Zhang, J. Han, H. Wang, R. Car, E. Weinan, Deep potential molecular dynamics: A scalable model with the accuracy of quantum mechanics. *Phys. Rev. Lett.* **120**, 143001 (2018).
82. A. P. Thompson, H. M. Aktulga, R. Berger, D. S. Bolintineanu, W. M. Brown, P. S. Crozier, P. J. in 't Veld, A. Kohlmeyer, S. G. Moore, T. D. Nguyen, R. Shan, M. J. Stevens, J.

- Tranchida, C. Trott, S. J. Plimpton, LAMMPS—A flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Comput. Phys. Commun.* **271**, 108171 (2022).
83. W. G. Hoover, B. L. Holian, Kinetic moments method for the canonical ensemble distribution. *Phys. Lett. A* **211**, 253–257 (1996).
84. R. M. Wentzcovitch, Invariant molecular-dynamics approach to structural phase transitions. *Phys. Rev. B* **44**, 2358–2361 (1991).
85. J. R. Ray, A. Rahman, Statistical ensembles and molecular dynamics studies of anisotropic solids. *J. Chem. Phys.* **80**, 4423–4428 (1984).
86. Y. Zhen, C. Chu, A deformation-fluctuation hybrid method for fast evaluation of elastic constants with many-body potentials. *Comput. Phys. Commun.* **183**, 261–265 (2012).
87. G. Clavier, A. P. Thompson, Computation of the thermal elastic constants for arbitrary manybody potentials in LAMMPS using the stress-fluctuation formalism. *Comput. Phys. Commun.* **286**, 108674–108674 (2023).
88. T. H. K. Barron, M. L. Klein, Second-order elastic constants of a solid under stress. *Proc. Phys. Soc.* **85**, 523–532 (1965).
89. C. Luo, J. Tromp, R. M. Wentzcovitch, Ab initio calculations of third-order elastic coefficients. *Phys. Rev. B* **106**, 214104 (2022).
90. S. Cottaar, T. Heister, I. Rose, C. Unterborn, BurnMan: A lower mantle mineral physics toolkit. *Geochem. Geophys. Geosyst.* **15**, 1164–1179 (2014).
91. J. Zeng, D. Zhang, D. Lu, P. Mo, Z. Li, Y. Chen, M. Rynik, L. A. Huang, Z. Li, S. Shi, Y. Wang, H. Ye, P. Tuo, J. Yang, Y. Ding, Y. Li, D. Tisi, Q. Zeng, H. Bao, Y. Xia, J. Huang, K. Muraoka, Y. Wang, J. Chang, F. Yuan, S. L. Bore, C. Cai, Y. Lin, B. Wang, J. Xu, J.-X. Zhu, C. Luo, Y. Zhang, R. E. A. Goodall, W. Liang, A. K. Singh, S. Yao, J. Zhang, R. Wentzcovitch, J. Han, J. Liu, W. Jia, D. M. York, W. E. R. Car, L. Zhang, H. Wang,

- DeePMD-kit v2: A software package for deep potential models. *J. Chem. Phys.* **159**, 054801 (2023).
92. D. P. Kingma, J. Ba, Adam: A method for stochastic optimization. arXiv:1412.6980 [cs.LG] (2014).
93. H. Wang, L. Zhang, J. Han, W. E, DeePMD-kit: A deep learning package for many-body potential energy representation and molecular dynamics. *Comput. Phys. Commun.* **228**, 178–184 (2018).
94. Y. Zhang, H. Wang, W. Chen, J. Zeng, L. Zhang, H. Wang, W. E, DP-GEN: A concurrent learning platform for the generation of reliable deep learning based potential energy models. *Comput. Phys. Commun.* **253**, 107206 (2020).
95. J. Liu, G. Zhang, J. Wang, H. Zhang, Y. Han, Research on Cu-Sn machine learning interatomic potential with active learning strategy. *Comput. Mater. Sci.* **246**, 113450 (2025).
96. K. Kawai, T. Tsuchiya, P-V-T equation of state of cubic CaSiO₃ perovskite from first-principles computation. *J. Geophys. Res. Solid Earth* **119**, 2801–2809 (2014).
97. L. Stixrude, C. Lithgow-Bertelloni, Thermodynamics of mantle minerals – I. Physical properties. *Geophys. J. Int.* **162**, 610–632 (2005).
98. Y. Zhang, D. Zhao, M. Matsui, G. Guo, Equations of state of CaSiO₃ perovskite: A molecular dynamics study. *Phys. Chem. Miner.* **33**, 126–137 (2006).
99. M. Noguchi, T. Komabayashi, K. Hirose, Y. Ohishi, High-temperature compression experiments of CaSiO₃ perovskite to lowermost mantle conditions and its thermal equation of state. *Phys. Chem. Miner.* **40**, 81–91 (2013).
100. Y. Kudo, K. Hirose, M. Murakami, Y. Asahara, H. Ozawa, Y. Ohishi, N. Hirao, Sound velocity measurements of CaSiO₃ perovskite to 133GPa and implications for lowermost mantle seismic anomalies. *Earth Planet. Sci. Lett.* **349-350**, 1–7 (2012).

101. N. Sun, H. Bian, Y. Zhang, J. F. Lin, V. B. Prakapenka, Z. Mao, High-pressure experimental study of tetragonal CaSiO_3 -perovskite to 200 GPa. *Am. Mineral.* **107**, 110–115 (2022).